UNIVERSITY OF BAHRAIN
COLLEGE OF INFORMATION TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE
BSC IN SOFTWARE ENGINEERING

ITSE305: Software Engineering Project

ATM SYSTEM BASED ON FINGERPRINT

ZAINAB A.WAHED ALSHAIKH 202008194
ZAINAB ALI AHMED 202002970
BATOOL ABDULLA 202009982

2022/2023 - Second Semester

SOFTWARE ENGENEERING PROJECT

ATM System based on Fingerprint.

Phase-1

FEBRUARY 19, 2023 – MARCH 4, 2023

TABLE OF CONTENTS

Table of Figure 5
Table of Table 6
EXECUTIVE SUMMARY 7
OVERVIEW 7
VISION 7
STRATEGIC OBJECTIVE 7
1.0 Pre-Initiation and initiation 8
1.1 Pre-Initiation 8
1.2 initiation 11
1.2.1 Project Charter 11
1.2.2 Stakeholders register. 12
1.2.3 management strategy 13
1.2.4 Project Kick-Off 13
2. planning 15
2.1. Team contract 15
2.2. Project Scope Management 16
2.2.1. Software process model plan and justification 16
2.2.2. Scope Statement 16
2.2.3. Scope Baseline 18
.2.2.4 ..... Scope validation 21
2.3. statement of work 22
2.4. Project Time Management 24
2.4.1. Gant chart 24
2.4.2. Network diagram 25
2.4.3. Project schedule 26
2.5. Risk Management Plan 27
2.5.1. Risk registers 29
2.6. Project Communication Management 31
3. Executing 33
3.1. System requirements 33
3.1.1. Define Quality Attribute Scenario 33

3.2. Survey 45
3.3. Security 46
3.4. System analysis and design 47
3.4.1. Create flow chart diagram 47
3.4.2. Create use case diagram 48
3.4.3. Create sequences diagram 49
3.4.4. Create activity diagram 50
3.5. implementation and user interface 50
3.6. integration and testing 52
4.0 Monitoring and controlling 54
4.1 Progress report 54
4.2 Milestone report 55
4.3 Change request 56
5.0 closing 60
5.1 Lessons Learnt 60
5.2 Prototype 61
Appendixes A 63
Meeting minute 63

TABLE OF FIGURE

Figure 2-1 project work breakdown structure (WBS) 18
Figure 2-2 WBC Dictionary 19
Figure 2-3 Gant Chart A 24
Figure 2-4 Gant Chart B 24
Figure 2-5 Gant Chart C 25
Figure 2-6 Gant Chart D 25x
Figure 2-7 Network diagram 25
Figure 2-8 WBS 27
Figure 3-1 Context diagram 37
Figure 3-2 Models digram 42
Figure 3-3 Initial deployment diagram 44
Figure 3-4 Flow Chart diagram 47
Figure 3-5 use case diagram 48
Figure 3-6 Sequences Diagram 49
Figure 3-7 Activity Diagram 50
Figure 3-8 User Interface B 51
Figure 3-9 User Interface A 51
Figure 3-10 User Interface C 52
Figure 5-1 Prototype A 61
Figure 5-3 Prototype B 62
Figure 5-2 Prototype C 62

TABLE OF TABLE

Table 1 Business Case 8
Table 2 Project Charter 11
Table 3 Stakeholders Register 12
Table 4 Management Strategy 13
Table 5 Team Contract 15
Table 6 Scope Statement 16
Table 7 Statement of Work 22
Table 8 Risk Management Plan 28
Table 9 Risk Register 29
Table 10 Risk Register 30
Table 11 Project Communication Management 31
Table 12 Quality Attribute 33
Table 13 Architectural Concerns 34
Table 14 Constraints 34
Table 15 Primary Functionality 34
Table 16 Review Inputs 35
Table 17 Design Concepts 38
Table 18 Design Decision 40
Table 19 elements and their Main Roles 43
Table 20 The Elements and their Responsibilities 44
Table 21 Perform Analysis 45
Table 22 Milestone Report 55
Table 23 Required Approvals ..... **Error! Bookmark not defined.**

EXECUTIVE SUMMARY

Our goal is to use fingerprint-based ATMs (Automatic Teller Machines) to develop a far better security mechanism. A technique called biometrics may help to provide highly secure data that is specific to everyone based on their distinctive physical traits. By using the individual's fingerprint, iris, face, voice, hand geometry, or handwriting, among other biometric features, the person may be identified with absolute accuracy. Mag tape cards, physical keys, and smart cards are examples of tokens that are frequently taken, lost, duplicated, or disappeared. Passwords are frequently forgotten, copied, hacked, or unintentionally seen by a third party.

Recognition and authentication are the two main functions that a biometric system performs.

These days, fingerprint processing is generally acknowledged and is a biometric technology that is completely developed. It is also simple to design for a high level of protection and safety at the user's fingers.

It is simple to carry out and may need very little time and effort to get a person's fingerprints registered on an identifying device.

Currently, automatic self-banking has become quite popular thanks to its enduring benefit of customer assistance that is available around-the-clock. The use of ATMs (Automatic Teller Machines) can provide users with the necessary banknote handling. Yet during the past few years, financial crime has increased significantly. Many thieves disrupt ATM depots and illegally take the victim's master card and means of identification. When a person's credit card is lost, the password is stolen as well. The criminal takes advantage of the situation quickly, which can result in significant financial losses for victims. As it hits the major objective in the current financial condition, the manner of keeping up is the genuine distinction to the consumer.

OVERVIEW

VISION

Using the ATM provides the user with a safe and secure experience.

STRATEGIC OBJECTIVE

- Get the cooperation of other banks by gaining their confidence to participate as stakeholders.
- Spend money on quality control.

1.0 PRE-INITIATION AND INITIATION

1.1 PRE-INITIATION

Business Case

Table 1 Business Case

1.0 Introduction/ Background

The core business goal of this project is to facilitate the conduct of banking transactions through the ATM by using a fingerprint instead of a bank card. The CEO of the National Bank of Bahrain believes that such a system will help many people by reducing the risk of being robbed. It also facilitates using ATMs, as there is no need to remember a PIN or carry a card with you. The project targets all people who have bank accounts and use ATMs a lot or those with more than one bank account and always need to remember the PIN of their credit cards.

2.0 Business Objective

The system is intended to provide high security for ATM users in Bahrain by replacing the bank card with a fingerprint. As a result, the system will make using the ATM and performing banking transactions easier and faster. In addition, applying the system will reduce the number of thefts due to bank cards, making people more satisfied with the current situation.

3.0 Current Situation and Problem/Opportunity Statement

People are constantly exposed to the theft of their credit cards, which leads to the theft of their money, and the procedures for obtaining a new credit card may take a long time. In addition, people will always be worried because there is no guarantee that they will not be stolen again. The fingerprint system will save people a lot and facilitate withdrawals and deposits via ATMs. Using the fingerprint ensures that no one can steal their money. Also, they will not have to carry a credit card or memorize the PIN.

4.0 Critical Assumption and Constraints

Fingerprint Based ATM is a system where fingerprint of the user is used as authentication. The fingerprint minutiae features are different for each human being so the user can be identified uniquely. Instead of using ATM card Fingerprint based ATM is safer and secure. There is no worry of losing your ATM card and no need to carry ATM card in your wallet. You just must use your fingerprint to do any banking transaction. The user can withdraw money from his account. and can transfer money to various accounts. the system will be available in all the bank ATM's. The system automatically sends notification to the User about the balance available in his respective account and the transaction details. Also, the project manager must create an effective work environment.

5.0 Analysis of Options and Recommendation

1- The system must be implemented by high-level Languages and supports JAVA, JAVASCRIPT, HTML, PHP,C++.
2- The system must contain high safety and accuracy in recognizing the fingerprint.
3- The system should be connected with the National Bank of Bahrain system.
4- The project leader should distribute the tasks according to team members' skills.
5- Any delays in this project will require an increase in work hours for the team members.

6.0 Preliminary Project Requirements

1- Login: User will login to the system using his fingerprint.
2- Withdrawal of cash: User can withdraw cash by entering the amount he wants to withdraw.
3- Transfer of money: User can transfer cash to other accounts by entering the account number he wants to transfer.
4- View balance: User can view balance which is available in his respective account.
5- View transaction: User can view the last four transactions.
6- Availability: The system must be available 24/7.
7- Notification: The system should be linked with user number to notify the user with all the transactions to fully aware what happening to the user's income and outcome by SMS.
8- Backup: A restoration copy is kept in case a security breach or server error occurs. Therefore, preventing data loss.

7.0 Budget Estimate and Financial Analysis

The cost of the full project is estimated to be 50,000 BD. This estimate assumes that the project manager will work around 23 hours per week for three months and that other internal staff members would work a combined 38 hours per week. The help of the customer service staff would not be compensated. The hourly wage for a staff project manager is 20 BD. Due to the possibility that certain hours typically invoiced to clients may be required for this project, the hourly cost for the other project team members would be 30 BD. The first cost estimate also accounts for 5,000 BD for buying supplies of software and services. Maintenance expenses up to 5,000 BD per year are covered when the project is finished. Benefits are anticipated based on a decrease in the number of hours consultants spend looking up project management resources, suitable tools, and templates. The anticipated advantages also take into account a modest rise in earnings brought on by the additional business this project would bring in. The expected advantage would be 9,000 BD per year if each of the 300 consultants could save just 30 hours a year (less than one hour per week) and bill those hours to other projects that earn, on average, 5 BD per hour. Using prior profit data, the new project might improve business by just 1%, resulting in enhanced earnings of at least 3,000 BD annually. Hence, the total estimated benefits equal to 12,000 BD annually. The predicted expenses and benefits are summarized in Exhibit A, which also displays the estimated net present value (NPV), return on investment (ROI), and year of payback. It also includes a summary of the presumptions used in this preliminary financial analysis. The financial projections are all extremely positive. As specified by the sponsor, the projected repayment time is one year. The discounted ROI based on a three-year system life is an amazing 269 percent, and the NPV is 169,300 BD.

8.0 Schedule Estimate

The sponsor wants the project completed within three months, but the schedule has some flexibility. We also assume that the new system will have a useful life of at least ten years.

9.0 Potential Risks

There are several risks in this project:

- First, schedule and budget constraints, the project may exceed the budget, and there may be some delay in completing the project on time, although the team will work their best to deliver it on time and within budget.
- Second, lack of flexibility to identify the person in case of a cut or wound or when fingerprints are smudged.
- Third, the project may have some management risk because of the poor quality of communication among the team members and the ineffective communication.
- Finally, the project may have some limitations in continuous development.

10.0 Exhibits

Exhibit A: Financial Analysis

Discount rate	8%				
Assume the project is done in about 3 months	Year				
	0	1	2	3	Total
Costs	50,000	5,000	5,000	5,000	
Discount factor	1	0.93	0.86	0.79	
Discounted costs	50,000	4,650	4,287	3,969	62,885
Benefits	0	90,000	90,000	90,000	
Discount factor	1	0.93	0.86	0.79	
Discounted benefits	0	83,333	77,160	71,445	231,939
Discounted benefits - costs	50,000	79,050	73,100	67,150	269,300
Cumulative benefits - costs	50,000	2,532	55,029	169,300	←NPV
Payback in 1 year					
Discounted life cycle ROI	**269%**				
Assumptions					
Costs	# Hours				
PM (300 hours, 20/hour)	6,000				
Staff (500 hours, 30/hour)	15,000				
Outsourced software and services	5,000				
Total project costs (all applied in year 0)	50,000				
Benefits					
# Consultants	300				
Hours saved	30				
BD/hour profit	5				
Benefits from saving time	9,000				
Benefits from 1% increase in profits	3,000				
Total annual projected benefits	12,000				

1.2 INITIATION

1.2.1 Project Charter

Table 2 Project Charter

Project Title: ATM System based on Fingerprint	
Project Start Date: February 21st ,2023	**Projected Finish Date:** May 27th 2023
Budget Information: The firm has allocated **50,000** BHD for this project. Most costs for this project will be internal labor. An initial estimate provides a total of **20** hours per week.	
Project Manager: Batool Abdulla – 202009982@stu.uob.edu.bh	
Project Objectives: This project's main objectives are to create a distinctive brand that serves as an authorized character and to design a safer ATM infrastructure. When the customer places their finger on the ATM's biometric scanner, and if the finger coordinate is found, the customer's name appears on the screen. Every transaction is prohibited if, by chance, that Fingerprint coordinate is not found. The user will be able to pick from all relevant accounts on the screen if they are connected, saving them from having to use numerous ATMs to complete transactions if they have several bank accounts.	
Success Criteria: The project is distributed to at least 3 banks within 2 years of completion, in which the project will be finished and delivered on the scheduled date.	
Approach: - Waterfall method, It follows a linear, sequential process for completing tasks and phases, and one part of the project must be finished before moving on to the next. - Examine how and where fingerprints are currently used. - With the approval of all the stakeholders, establish and build an expected budget for the project's development. - comments and evaluations from experts on the project's potential. -	

Roles and Responsibilities			
Name and Signature	***Role***	***Position***	***Contact Information***
National Bank of Bahrain (NBB)	***sponsor***	***CEO***	*17214433*
Dr. Fawzi Alblooshi	Supervisor	Project advisor	falblooshi@uob.edu.bh
Zainab Alshaikh	Team member	Designer	202008194@stu.uob.edu.bh
Zainab Ali	Team member	Programmer	202002970@stu.uob.edu.bh
Batool Abdulla	Team member	Project Manger	202009982@stu.uob.edu.bh
Ms. Afrah Ali	Supervisor	Evaluator	Ayusuf@uob.edu.bh
Sign-Off : *Fawzi* *Zainab* *Zainab Ali* *Batool* *Afrah*			
Comments: If anyone has a wonder, do not hesitate to ask. - Batool Abdulla			

1.2.2 Stakeholders register.

Table 3 Stakeholders Register

Name	**Role**	**Position**	**Internal / External**	**Contact Information**
National Bank of Bahrain (NBB)	sponsor	CEO	External	*17214433*
Dr. Fawzi Alblooshi	Supervisor	Project advisor	External	falblooshi@uob.edu.bh
Zainab Alshaikh	Team member	Designer	Internal	202008194@stu.uob.edu.bh
Zainab Ali	Team member	Programmer	Internal	202002970@stu.uob.edu.bh
Batool Abdulla	Team member	Project Manager	Internal	202009982@stu.uob.edu.bh
Ms. Afrah Ali	Supervisor	Evaluator	Internal	Ayusuf@uob.edu.bh

1.2.3 management strategy

Table 4 Management Strategy

Name	Level of Interest	Level of Influence	Potential Management Strategies
National Bank of Bahrain (NBB)	High	High	Being the CEO and the first bank to implement fingerprint identification, NBB is the project's main key.
Batool Abdulla	High	High	Along with the team, the project manager creates the project plan and keeps tabs on how the team is doing with the project's objectives and makes sure it is accurately done according to the plan.

1.2.4 Project Kick-Off

Kick-Off Meeting

February 15, 2023, 1:00pm – 3:00pm

Project Name: ATM System based on Fingerprint.

Meeting Objective: Selecting the topic of the project and discussing the main goals and functionalities. Also, to determine the roles and responsibilities of each stakeholder.

Agenda:

1. Introduce all stakeholders.
2. Choosing the topic of the project.
3. Determine the main goals and functionalities of the project.
4. Determine the roles and positions of each team member.
5. Review the project document.
6. Distributing the work among the team members.
7. Decide the next meeting date and time.

Action Item	Assigned To	Due Date
Business case	All team member	March 4, 2023
Stakeholder register	All team member	February 22, 2023
Stakeholder strategy	All team member	February 22, 2023
Project charter	Batool Abdulla	March 1, 2023

Date and time of next meeting: February 22, 2023, 1:00 pm – 3:00 pm

---

SOFTWARE ENGENEERING PROJECT

ATM System based on Fingerprint.

Phase-2

MARCH 5, 2023 – APRIL 15, 2023

2. PLANNING

2.1. TEAM CONTRACT

Table 5 Team Contract

Code of conduct: As a project team, we will:

- Keep good and consistent contact between all team members to stay informed on the project.
- Be honest and realistic in planning and reporting project scope and schedule.
- Decisions are made based on the general interest of all project members
- The team will work by anticipating potential issues and attempting to forestall them.

Participation: We will:

- Maintain a high level of credibility and transparency throughout the project duration.
- Encourage a diversity of opinions on all topics.
- Welcome new ideas and accept new approaches to work.
- Give equal opportunity for team members to participate.
- Keep the project manager aware of the team member attendance at the meetings.
- Provide solutions that satisfy all parties.

Communication: We will:

- The meetings will organize face-to-face, and the team will use email to send important documents. Also, WhatsApp will be for basic notifications, such as appointment scheduling.
- Keep ideas consistent and not contradictory to each other.
- Be clear and to the point.
- Keep discussions on track.
- After collecting and merging all the work for each phase, all team members view and read the whole document to suggest or edit anything if accepted.

Problem solving: We will:

- The project manager is keen to encourage all members to participate in finding possible solutions to the problems.
- Avoid blame and frustration and focus on the problem to find a solution and avoid it in the future.
- Consider all ideas and suggestions to build a perfect solution.

Meeting guidelines: We will:

- Meetings will be held in room TE2023 every Sunday and Thursday. In addition, a meeting will be held whenever it is needed.
- Zainab Ali undertakes to note all decisions and actions that are made.

2.2. PROJECT SCOPE MANAGEMENT

Project scope management is a process that helps in determining and documenting the list of all the project goals, tasks, deliverables, schedules, and budgets as a part of the planning process.

2.2.1. Software process model plan and justification

For our project, we discussed the possible processes to follow and the team members decided that the best software process model to use is the waterfall model for various reasons. First, it is one of the easiest and simplest processes for software projects. Also, the requirements for the project are all precise, and there is no ambiguity since the main source of the requirements is the team members responsible for the project.

2.2.2. Scope Statement

Table 6 Scope Statement

Project Title: ATM system based on Fingerprint**.** **Date:** 7 March 2023 **Prepared by:** Batool Abdulla, project manager, 202009982@stu.uob.edu.bh
Project Summary and Justification: This initiative aims to make it easier to execute financial transactions at ATMs by utilizing a fingerprint rather than a bank card. Our sponsor NBB thinks that by lowering the chance of robbery, such a system will benefit many individuals. It also makes it easier to use ATMs because you don't have to carry a card around or remember a PIN. The team members were adamant about offering a simple, efficient method. The project has a 50,000 BHD budget.
Product Characteristics and Requirements: **1-** Design: The system will be easy to understand and retain for all users, whether they are experts in systems or novices. The user interface will be uncluttered and clean, and it will be simple to navigate from the homepage to the needed capabilities. The selection of colors, images, and layouts will take user preferences into consideration. **2-** Security: One of the sensitive pieces of information that cannot be read or changed by anybody is a user account. As a result, a licensed API will be employed to uphold a high degree of security. **3-** Performance: The reaction time and number of clicks will be kept to a minimum while the productivity of the user is taken into account. A large number of users must be supported by the system without any issues. **4-** Accessibility: Users will have constant access to the system. Users may always access the system to complete their transactions. The system will be constructed in a maintainable fashion that helps users constantly and does not prevent them from doing their activities in order to satisfy the changing requirements of stakeholders and new technologies. **5-** Database: Users may always examine their most recent four transactions in the database. In addition to having access to monthly statements **6-** Notification: In order to properly inform the user of all transactions and their impact on their income and results, the system should be connected to the user's number. **7-** Access: The user will use his fingerprint to sign into the system. to monitor balance and transactions, withdraw money from an account, and transfer money

8- Backup: In the event of a security breach or server mistake, a restore copy is preserved. avoiding data loss as a result.

Summary of Project Deliverables

Project management-related deliverables: Business case, charter, team contract, scope statement, WBS, schedule, cost baseline, progress reports, final project presentation, final project report, lessons-learned report, and any other documents required to manage the project.

Product-related deliverables:

1- Find database programs that can handle big amounts of data.
2- Survey: To ascertain the features required for a positive banking experience. All bank customers will be included in the survey to provide input on the system's concept.
3- Using an iterative development process can help gather feedback, address any issues, and boost productivity.
4- Promotion: Promotion will draw attention to the system's launch.
5- system design: The suggested layouts, colors, and graphics will be used at first. User comments about the preliminary design will be used to inform the final design.
6- Test strategy: The system will be put through four stages of testing, starting with unit testing individual components, moving on to integration testing integrated components, system testing the complete system, and acceptance testing the finished system.
7- Project management tools: Using MS Project for the WBS, Gantt charts, network diagrams, cost estimates, and earned value management.

Project Success Criteria

Members of the team are certain that the application must be delivered in three months while staying within the 50,000BHD budget. After the system has been delivered and tested, we want to hear user feedback because the system's success depends on it.

2.2.3. Scope Baseline

Work breakdown Structure (WBS)

- **1.** Pre-Initiation and initiation
  - **1.1.** Pre-Initiation
    - **1.1.1.** Prepare business case.
  - **1.2.** Initiation
    - **1.2.1.** Identify key stakeholders
    - **1.2.2.** Prepare project character
    - **1.2.3.** Document stakeholder register
    - **1.2.4.** Stakeholder management strategy
    - **1.2.5.** Hold project kick-off meeting
- **2.** Planning
  - **2.1.** Prepare team contract
  - **2.2.** Prepare scope statement
  - **2.3.** Scope baseline
    - **2.3.1.** Prepare WBS and WBS dictionary
  - **2.4.** Prepare project schedule
    - **2.4.1.** Determine task duration
    - **2.4.2.** Determine task dependencies
    - **2.4.3.** Create draft Gantt chart
    - **2.4.4.** Review and finalize Gant chart
    - **2.4.5.** Create network diagram
  - **2.5.** Risk management plan
    - **2.5.1.** Identify, discuss, and prioritize risks
- **3.** Executing
  - **3.1.** System requirements
    - **3.1.1.** Define quality attribute scenario
    - **3.1.2.** Define Functionality requirements
  - **3.2.** Survey
  - **3.3.** Security
  - **3.4.** System analysis and design
    - **3.4.1.** Create flow chart diagram
    - **3.4.2.** Create use case diagram
    - **3.4.3.** Create sequences diagram
    - **3.4.4.** Create activity diagram
  - **3.5.** implementation and user interface
  - **3.6.** integration and testing
- **4.** Monitoring and controlling
  - **4.1.** Progress report
- **5.** Closing
  - **5.1.** Prepare final report.
  - **5.2.** Lessons learned

Figure 2-1 project work breakdown structure (WBS)

WBS Dictionary

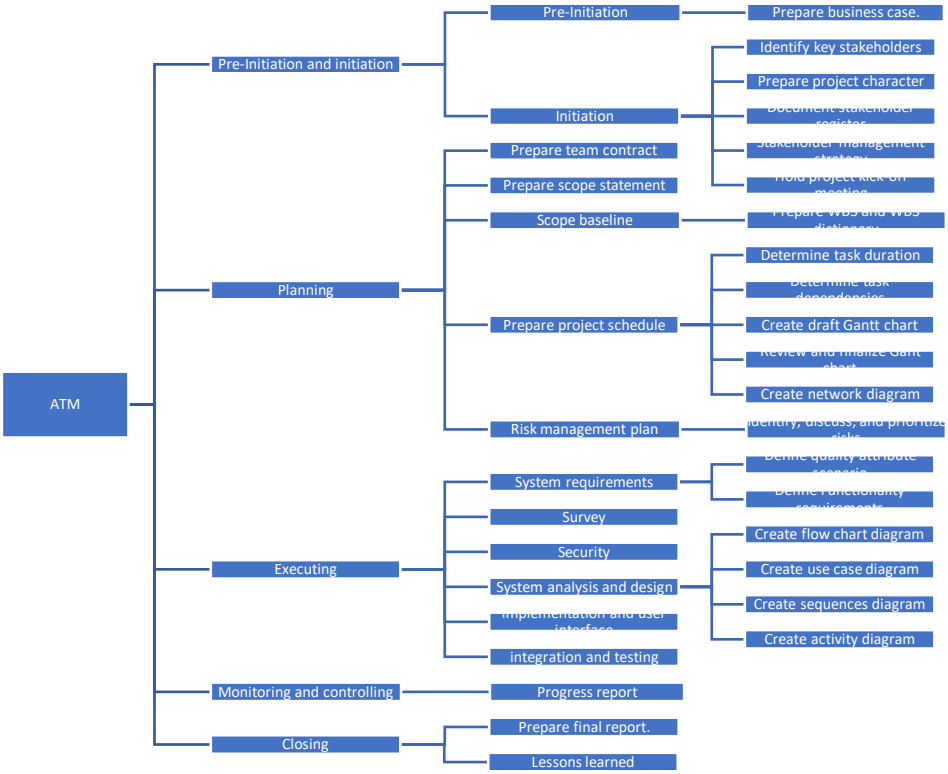

Figure 2-2 WBC Dictionary

Prepare business case: document with the proposed project highlighting the benefit from it.

Identify key stakeholders : people affected by your business who do not directly work with you.

Prepare project character : A document for the approved project and approach to manage it and the acceptance criteria.

Document stakeholder register : A table with all the stakeholders involved in the development process.

Stakeholder management strategy: A plan of communication with all stakeholders.

Hold project kick-off meeting: A table with the information regarding the first meeting held.

Prepare team contract : A document setting the values and ethics the team members will follow.

Prepare scope statement : A document briefly discussing the requirements and the scope of the project.

Scope baseline : The approved version of a scope statement, work breakdown structure (WBS), and its associated WBS dictionary.

Prepare WBS and WBS dictionary : A plan that divides the project into smaller components. And document providing additional information about the parts of the project.

Prepare project schedule : A detailed plan with all the tasks and activities with their durations and dependencies.

Determine task duration : dividing the number of hours it actually takes you to complete a task by the number of hours you thought it would take.

Determine task dependencies : analyzing the start and finish dates of the dependent tasks.

Create draft Gantt chart : A chart used to visually represent the schedule with dependencies and durations.

Review and finalize Gant chart : make sure it is well done chart.

Create network diagram : A Network Diagram is a graphical way to view tasks, dependencies, and the critical path of the project.

Risk management plan : A plan highlighting the people wo will deal with the risks.

System requirements : The system analyst is responsible for producing from the stakeholders a list of criteria that will become the basis of the specifications of the project.

Define quality attribute scenario : A list of the nonfunctional requirements for the system.

Define Functionality requirements : A list of all the functionalities of the system with the scenarios.

Survey : provider of reliable, usable, primary data to inform business decisions.

Security : the quality or state of being secure

System analysis and design : An explanation of what design process will be chosen to follow.

Create flow chart diagram : The diagrams that show the behavior of the systems functionalities.

Create use case diagram : Diagrams to show interaction between users and the system.

Create sequences diagram : Diagrams to show the sequencing of events when a functionality is used.

Create activity diagram : A diagram visually presents a series of actions or flow of control in a system.

implementation and user interface : Aims to create interfaces which users find easy to use and pleasurable. The manager is liable for growing an implementation plan primarily based totally at the standards installed with the aid of using high education, the implementation plan must encompass the schooling wished for the software, support.

integration and testing : Integration plan and testing.

Progress report : Regular checkups are necessary for the system, as well as the progress reports which are created by the project manager and it tells the stakeholders in which stage exactly the project is now, if there are delays or not and what resources are needed. In addition, the manager should document the status of the system occasionally. If any fault were found, the manager should write a report and send it to the system designer, or programmer.

Prepare final report : Finalized the report: .

Lessons learned : A reflective report of the lessons learned through making the project.

2.2.4.Scope validation

Validating the scoping plan is one of the important plans that must be built in the project planning phase. It is the process of formalizing acceptance of the completed project deliverables. Validating the scoping plan is important in the project's success, mainly showing the stakeholders have received what was agreed upon and formalizing their approval. In addition, it is mainly concerned with recognizing the product by validating each main deliverable in the project.

The purpose of this plan in our project is to receive formal acceptance of our project's products and procedures to evaluate whether they were completed correctly and satisfactorily to meet the project requirements and our customer needs.

Table 7 Scope Validation

The main inputs of scope validation:

- The scope management plan
- Scope baseline
- Requirements documentation
- Requirements traceability matrix
- Validated deliverables

The main tools for performing scope validation:

- Inspection
- Group decision-making techniques

The main outputs of scope validation:

- Accepted deliverables
- Work performance information

Deliverables Acceptance:

The project deliverables will be formally presented and reviewed by team members. Then they will be accepted or rejected by the project manager and NBB during the monitoring and controlling phase. Anyone from the team members can present.

Name	Responsibility
All team members	Presents the project deliverables
All team members	Reviews the project deliverables
Batool	Verifies the project deliverables
Batool and NBB	Accepts or rejects the project deliverables

Accepted Deliverables:

After presenting, reviewing, and verifying the project deliverables, the final acceptance decision goes to project manager Batool and the client NBB.
The accepted deliverables are available in the scope statement.

2.3. STATEMENT OF WORK

A statement of work is a document that describes the products or services to be created by the project team. It usually includes a description of the business need for the project, a summary of the requirements and characteristics of the products or services, and organizational information, such as appropriate parts of the strategic plan, showing the alignment of the project with strategic goals. It also summarizes project requirements, individual tasks, and to whom these tasks are assigned.

Table 8 Statement of Work

Project title: ATM System Based on Fingerprint	
Date: March 24th, 2023	**Prepared by:** Zainab Ai and Zainab Alshaikh

Scope of Work:
The work is divided into 5 phases:

- **Pre-initiation and initiation phase:**
  - **Pre-initiation:**
    1. Business Case: must be initialized at the beginning of the project, it includes business objectives, scope statement, opportunity statement, critical assumption and constraints, recommendations, Preliminary Project Requirements, and potential risks.
  - **Initiation:**
    1. Project Charter: recognizes the importance of the project and puts down the project objectives and management.
    2. Stakeholders Register: defines each project client/team member, role, and responsibilities, along with his contact information.
    3. Stakeholders Management Strategy: identifies and documents the approach to take in order to increase support and decrease negative impacts of stakeholders throughout the life of the project.
    4. Project Kick-off Meeting: identifies of the essential elements of the project, goals of the project, and other project planning activities.

- **Planning phase:**
  1. Team Contract: describes all the commitments that the team members are willing to stand by during the project.
  2. Project Scope Management: consist**s** of planning the software process model, collecting requirements plan, scope statement which details the project deliverables and describes the major objectives, scope baseline, and scope validation. The scope baseline includes both WBS and WBS dictionary.
  3. Statement of Work: illustrates project-specific activities, deliverables and timelines for a vendor providing services to the client.
  4. Project Time Management: keeps track of time and resources using previous tasks for the sake of maki**ng** efficient use of the project's resources and deliver**ing** the project as quickly as possible. Project schedule, Gantt char, and network diagram will be established here.
  5. Risk Management Plan: identifies the risks that may be encountered throughout the project lifecycle and find ways to repudiate them.
  6. Project Communication Management: defines the frequency and the way of communication among the project stakeholders.

- **Execution phase:**
  1. Requirements Management: elicit functional and non-functional requirements and analyze, validate, generate software requirements specifications, and verify. A prototype will be designed to gain an understanding and determine the requirements' feasibility. After that, we will generate a requirements traceability matrix to ensure that all requirements are addressed. Then, the user characteristics will be defined with the principal stakeholders, concerns, and constraints.
  2. Design Process: The system will be decomposed into smaller parts to ensure ease of design for each component separately, and then the components will be integrated.
  3. Milestone Reports: will be performed to ensure that all of the project processes are on track and completed on time.
  4. Performance Reports: will be provided for stakeholders as status reports or progress reports to keep them on track.

- **Monitoring and controlling phase:**
  1. Team Performance Assessment: reports the assessment of the team members in the entire project.
  2. Change Request is a document containing a call to adjust a system; it is important in the change management process.
  3. Milestone Reports will be performed to ensure that all of the project processes are on track and completed on time and take corrective action when necessary.
  4. Performance Reports will be provided for stakeholders as status reports or progress reports to inform them of progress made.
- **Closing phase:**
  1. Lesson-Learned Report documents every important event, challenge, constraint, risk, and uncertainty faced during the project.
  2. Final project Report Table of Contents lists all project contents from pre-initiation to clothe sing phase.

Location of Work:
Team members will work at the company, and meetings will also be held there.

Period of Performance:
The services of this system will be required for a period of approximately three months, starting from February 21st,2023. The expected completion date of this project is May 27th, 2023.

Deliverables Schedule:

Week	Project Activity
0-2	**Pre-initiating and Initiating**
2-5	**Planning**
5-9	**Execution**
9-11	**Monitoring and Controlling**
11-12	**Closing**

Applicable Standards:
The deliverable system should satisfy The National Bank of Bahrain NBB goals.

Acceptance Criteria:

All deliverables are to be inspected by the NBB and revised by team members, if anything was not satisfactory, they have the authority to reject it or require correction.
Special Requirements All the team members: should have experience with web development, system analysis, database, and design.

2.4. PROJECT TIME MANAGEMENT

2.4.1.Gant chart

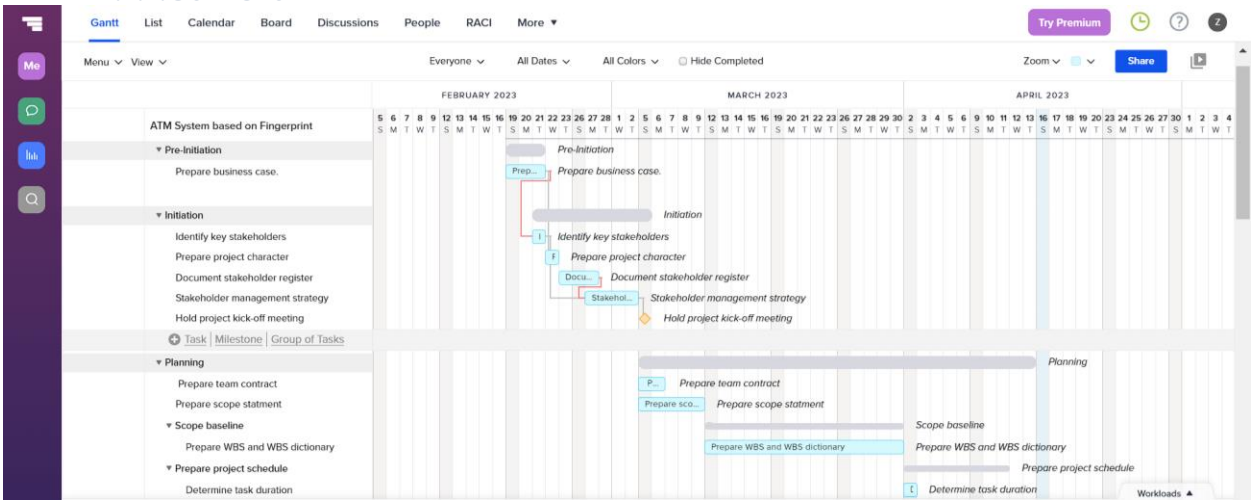

Figure 2-3 Gant Chart A

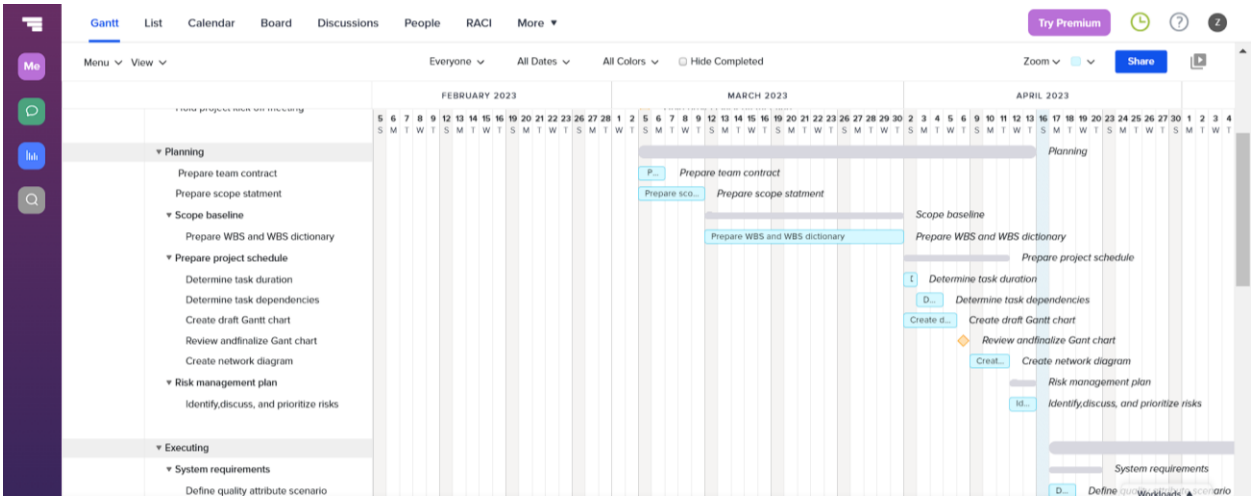

Figure 2-4 Gant Chart B

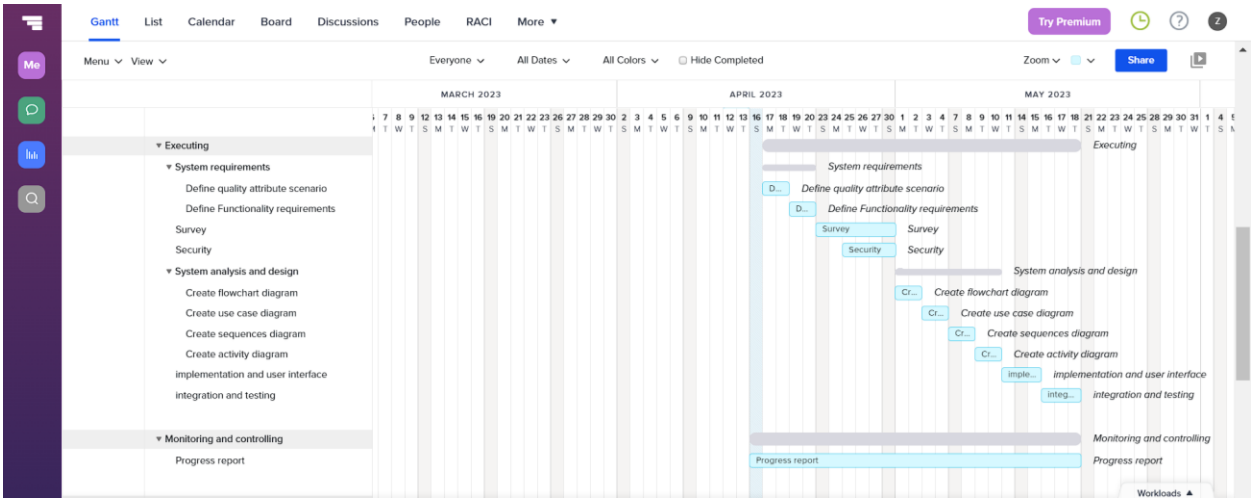

Figure 2-5 Gant Chart C

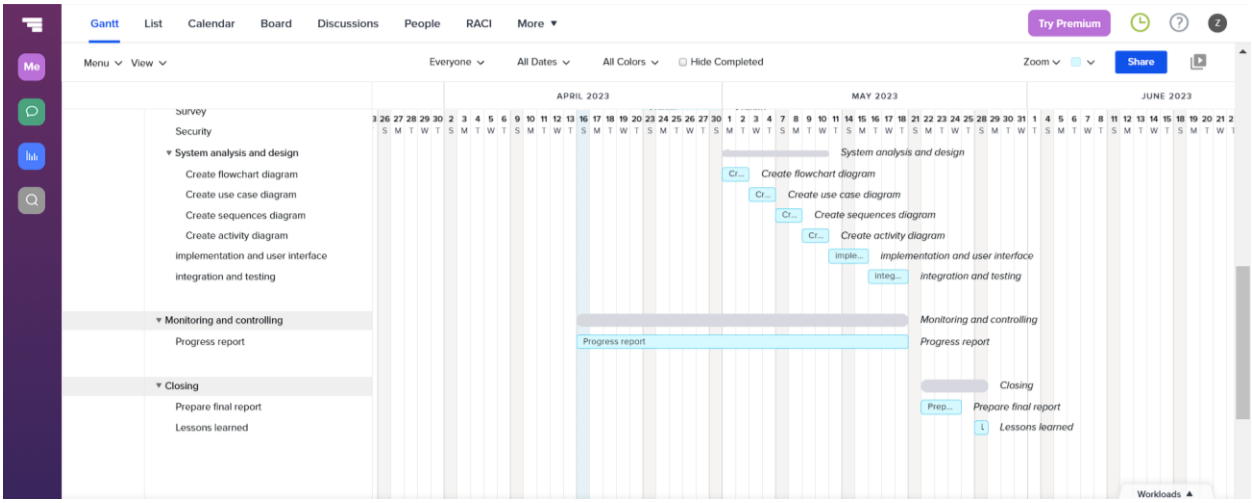

Figure 2-6 Gant Chart D

2.4.2.Network diagram

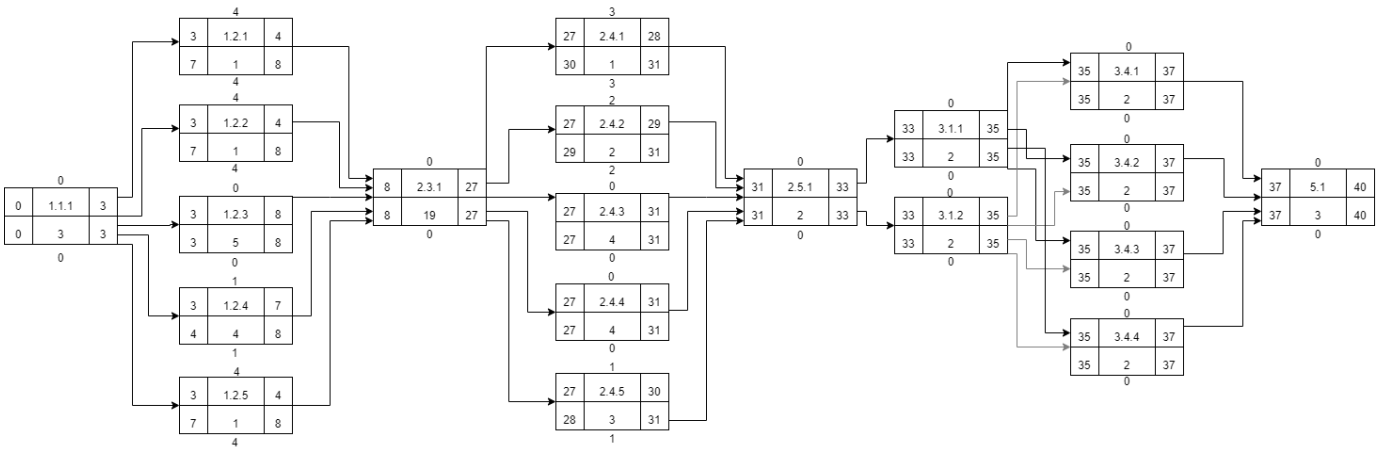

Figure 2-7 Network diagram

2.4.3.Project schedule

In project scheduling, a bunch of duties, tasks. And activities are broken down into a set of five phases which are circulated and assigned to each member of the team. The five phases are: pre-initiation and initiation, planning, executing, monitoring and controlling, and closing. Each phase includes several tasks, and they are needed to be delivered under a given of time.

Pre-initiation: The idea of the system was chosen in the brainstorming session in a meeting that was held and includes all the team members, also the project sponsor was opted. In addition, the business case, schedule, and cost were developed, and all have been discussed and approved by the sponsor as well as the supervisor.

Initiation: A kick-off meeting was held between the team members, roles and responsibilities were distributed between them. Moreover, the project charter was developed. Beside the registry of the stakeholders, a stakeholder's management strategy was created.

Planning: A team contract was formed, project scope management was established with the statement of work, along with that project time management was done with the estimation of the project schedule. Furthermore, project communication management and risk management were developed in this phase, also the project work plan was created.

Executing: There are three main stages in this phase which are: system requirements, design process, and application prototype

Monitoring and Controlling: Stakeholders and team members will hold a meeting together to make a full checkup on the system and examine it, as well as reviewing the system to present a progress report. Also, changing and modifying requests may occur in this phase.

Closing phase: All deliverables are finalized and approved, the final project report will be submitted, a final project presentation will be ready to present.

ATM System based on Finge...	start	end
Pre-Initiation	**02/19/23**	**02/21/23**
Prepare business case.	02/19	02/21
Initiation	**02/21/23**	**03/05/23**
Identify key stakeholders	02/21	02/21
Prepare project character	02/22	02/22
Document stakeholder register	02/23	02/27
Stakeholder management strategy	02/27	03/02
Hold project kick-off meeting	03/05	03/05
Planning	**03/05/23**	**04/13/23**
Prepare team contract	03/05	03/06
Prepare scope statment	03/05	03/09
Scope baseline	**03/12/23**	**03/30/23**
Prepare WBS and WBS dictionary	03/12	03/30
Prepare project schedule	**04/02/23**	**04/11/23**
Determine task duration	04/02	04/02
Determine task dependencies	04/03	04/04
Create draft Gantt chart	04/02	04/05
Review andfinalize Gant chart	04/06	04/06
Create network diagram	04/09	04/11
Risk management plan	**04/12/23**	**04/13/23**
Identify,discuss, and prioritize risks	04/12	04/13
Executing	**04/17/23**	**05/18/23**
System requirements	**04/17/23**	**04/20/23**
Define quality attribute scenario	04/17	04/18
Define Functionality requirements	04/19	04/20
Survey	04/23	04/30
Security	04/25	04/30
System analysis and design	**05/01/23**	**05/10/23**
Create flowchart diagram	05/01	05/02
Create use case diagram	05/03	05/04
Create sequences diagram	05/07	05/08
Create activity diagram	05/09	05/10
implementation and user interface	05/11	05/15
integration and testing	05/16	05/18
Monitoring and controlling	**04/16/23**	**05/18/23**
Progress report	04/16	05/18
Closing	**05/22/23**	**05/28/23**
Prepare final report	05/22	05/24
Lessons learned	05/28	05/28

Figure 2-8 WBS

2.5. RISK MANAGEMENT PLAN

Risk management is a systematic process of analyzing and responding to project risks. It includes the probability and consequences of positive events and the probability and consequences of negative events to project objectives. Project risks are discussed with stakeholders and evaluated, and the causes of these risks are referred to. After discussing the risks, a plan is defined for their management and a strategy showing how the project team will deal with them to achieve the right goal.

Methodology: Risks may be found using a variety of methods and techniques, including root cause analysis, checklist analysis, brainstorming, and SWOT (Strengths, Weakness, Opportunities, and Threats) analysis.

Table 9 Risk Management Plan

Roles and Responsibilities:	
Risk Manager	- Updating and maintaining the Risk Management Database - Maintaining this Risk Management Plan - Informing the group of dangers' current state -monitoring progress toward bringing moderate and high risk to manageable levels -providing training in risk management -facilitating risk assessments - Writing reports, documentation, and risk briefings needed for project reviews.
Project Sponsor	- Examine and suggest to the risk manager modifications to the overall risk management strategy in light of the lessons gained. - Participate in the updating of program risk assessments made during the preceding quarter every quarter, or as instructed. - Examine the risk assessments conducted and the suggested risk mitigation measures, and suggest any adjustments that should be made. -Email the risk manager to notify them of any new threats. - Complete the specified mitigation tasks and notify the project manager of their status and completion so that they may be entered into the database. - Take part in risk assessment.
Project Manager	- Take part in risk identification. - Verify that everyone involved in the project is actively engaged. - Assure a thorough and recorded evaluation of the procedures and results of risk management. - Keep an eye on outside developments and alterations that are connected to the hazards. - Assign the risk triggers and priority level to each risk.
Project Stakeholders	- carries out the steps and actions linked to risk management. - Attend the risk manager's weekly meetings.
Team Members	- Attend the risk manager's weekly meetings. - Implement a framework for risk management and risk assessment. - Track risky behaviors
End Users	- The project will involve the end users. Risks may be identified by the end users, who should advise the project team about them. - The project team member(s) assigned to the End User will be responsible for identifying risks, assigning tasks to them, and reporting on them.

Risk probability and impact:

The risk management plan goes through many procedures that are carried out by risk managers with the assistance of the project managers and the involvement of the project participants. First, the risks are determined by the risk manager and the project manager by examining what may go wrong and what possible issues there might be while taking into account the environmental hazards and the fact that the environment is always changing. Measuring the frequency and impact of the risk, should it materialize, is a crucial step. Risk also affects terminology and categories. Each event's description and amount of effect state are given, either with a number value or a plain language phrase. Three states—low, moderate, and high—can be seen. The risk manager will understand the effects of each risk if the risk probability of occurrence is also

determined within a certain range. The management will have the ability to analyze the possible hazards, list them out, and set up a routine risk assessment.

The team members will examine the risks while focusing on the project's life cycle, processes, and procedures, while also acknowledging the impact risks and likelihood of occurrence that are crucial for creating the plan. The risk identifier will do a root cause analysis using the techniques currently in use while taking into account the circumstances that lead to the determination of the risk, how to address it, and how to prevent a recurrence.

Tracking and Monitoring Risks: Tracking and Monitoring Risks entails looking for identified, residual, and secondary risks as well as identifying any new risks. It also entails taking swift corrective action when a risk manifests, planning additional preventive measures when a new risk trend is discovered, and gauging the success of risk responses. The risk manager will update the information and hazards' current status. The risk manager decides on auditing at meetings with the project team members.

Risk Documentation: This action follows risk identification since it allows the risk identifier to separate everything linked to the risks, including their sources, classifications, analyses, chances that they will occur, their effects, their timelines, and more. Therefore, a plan of action will be created while taking into account its limitations, sensitivity, and the project's risk tolerance.

2.5.1.Risk registers

A document and tool in risk management and project management is the risk register. It is used to identify possible risks in a project or organization, occasionally to comply with regulatory requirements, but primarily to keep track of potential problems that can thwart planned goals. The following phase in risk management is strategically attempting to control the possible problems. The risk registry contains all relevant information about each risk that has been discovered, including its type, degree, owner, and any mitigation measures that have been put in place to address it. It is beneficial to have a system in place to gather prospective risks and then chart a course for getting the project back on track should those risks materialize.

Risk Register PART1:

Table 10 Risk Register

ID No	Rank	Risk	Category	Root Cause	Triggers
R1	1	Low productivity	Poor Productivity	extended project schedules and a feeling of urgency	Members in the group experience Parkinson's Law.
R2	2	Breakdown of Specifications	Requirement Management	Coding and integration reveal that the specification is lacking or has needs that are in conflict.	Documentarians' requirements have not been properly revised Unresolved lack of consistency in the criteria
R3	3	Budget Adjustments	Cost Management	Requirements and scope are modified	-New requirements -incorrect comprehension of the necessary functionalities

R4	4	Requirements Inflation	Requirement Management	In later phases, new requirements and functionality will be identified.	As the project moves forward, a growing number of elements that were not discovered at the project's commencement endanger estimates and timeframes.

Risk Register PART2:

Table 11 Risk Register

ID No	Potential Responses	Risk Owner	Probability of Occurrence	Impact	Status
R1	a meeting with the software analysts and the entire team	Project Manager	Medium	High	Short iterations, assembling the ideal team, coaching, and developing the team.
R2	Constant documentation review and repeated project manager oversight of software analysts	Software Analyst	Medium	High	Use a specialized product manager to decide on important trade-offs. Resolve assumptions and choices that disagree.
R3	the risk manager choosing the best strategy for a specific budget	Project Manager	Medium	Medium	Call a meeting and make sure everyone on the team is aware of the criteria.
R4	At the end of each iteration, the project manager and software analyst have routine trade-off conversations concerning features and budgets.	Software Analyst	Low	High	ongoing interaction between clients and developers

2.6. PROJECT COMMUNICATION MANAGEMENT

Teamwork and stakeholder involvement are the base for any project to succeed, and communication is the golden key to comprehensive and effective relationships between members. And this implies in every type of project, including software projects, thus, defining an effective and efficient approach for communication between the team member and stakeholders is considered a substantial element to let all members have a comprehensive understanding of the project and participate in it is formation. The meetings will be organized face-to-face, and the team will use email to send important documents. Also, WhatsApp will be for basic notifications.

Table 12 Project Communication Management

Name	Role	Position	Frequency	Tools Used
National Bank of Bahrain (NBB)	sponsor	CEO	Once a month	Email
Dr. Fawzi Alblooshi	Supervisor	Project advisor	Once a week issues discussion.	MS teams
Zainab Alshaikh	Team member	Designer	Twice a week discussion, work meetings, and exchange of messages and files.	Email and WhatsApp
Zainab Ali	Team member	Programmer	Twice a week discussion, work meetings, and exchange of messages and files.	Email and WhatsApp
Batool Abdulla	Team member	Project Manager	Twice a week discussion, work meetings, and exchange of messages and files.	Email and WhatsApp
Ms. Afrah Ali	Supervisor	Evaluator	Once a week issues discussion.	Email and MS teams

SOFTWARE ENGENEERING PROJECT

ATM System based on Fingerprint.

Phase-3

APRIL 16, 2023 – MAY 20, 2023

3. EXECUTING

3.1. SYSTEM REQUIREMENTS

3.1.1. Define Quality Attribute Scenario

Table 13 Quality Attribute

ID	Quality Attribute	Scenario	Related UC
QA-1	Security	An unauthorized user attempts to access an ATM using a fake fingerprint. The system detects the fake fingerprint and denies access. The system also alerts the bank and records the attempt. The response measure is the percentage of fake fingerprints that are detected and rejected by the system.	UC-1 UC-2
QA-2	Accuracy	A registered user tries to access an ATM using their fingerprint. The system matches the fingerprint with the stored biometric data and grants access. The system also displays the correct account information and balance. The response measure is the percentage of correct matches and mismatches between fingerprints and biometric data.	UC-2 UC-5
QA-3	Usability	A new user wants to register their fingerprint with the system. The system guides the user through the registration process and captures their fingerprint image. The system also verifies the quality of the image and asks the user to repeat the process if necessary. The response measure is the time and effort required for the user to complete the registration process.	UC-1 UC-5
QA-4	Reliability	A power outage occurs while a user is performing a transaction at an ATM. The system saves the transaction data and resumes the operation when the power is restored. The system also notifies the user about the status of the transaction. The response measure is the percentage of transactions that are completed or recovered after a power outage.	UC-5 UC-6
QA-5	Scalability	The number of users and transactions increases significantly over time. The system maintains its performance and functionality under the increased load. The system also adapts to the changing demand and allocates resources accordingly. The response measure is the throughput and latency of the system under different load conditions.	UC-1 UC-2 UC-3 UC-4 UC-5 UC-6
QA-6	Maintainability	A new feature or requirement is introduced to the system. The system can be modified or extended to accommodate the change with minimal impact on the existing functionality and quality. The system also follows coding standards and documentation practices to facilitate the change. The response measure is the effort and time required to implement and test the change.	UC-1 UC-2 UC-3 UC-4 UC-5 UC-6

3.1.2.List Architectural Concerns

Table 14 Architectural Concerns

ID	Concerns
CON-1	Establishing an overall initial system structure.
CON-2	Ensure the accuracy and reliability of the fingerprint sensors and scanners, especially in different environmental conditions and with different types of users.
CON-3	Protect the fingerprint data from being stolen or tampered by malicious actors, such as through skimming, hacking, or biometric forgery.
CON-4	Balance the trade-off between security and convenience for the users, such as by providing alternative authentication methods in case of fingerprint failure or user preference.
CON-5	Maintain the cost-effectiveness and profitability of the fingerprint-based ATMs, considering the additional expenses of installing and operating the biometric sensors and scanners.
CON-6	Arabic and English are the languages that this system uses.

3.1.3.Constraints

Table 15 Constraints

ID	Constraints
CONS-1	The scalability and adaptability of the fingerprint-based ATMs, which might require updating or replacing the biometric sensors and scanners as the technology evolves or new threats emerge.
CONS-2	The user acceptance and satisfaction of the fingerprint-based ATMs, which might depend on the ease of use, speed, and accuracy of the biometric authentication.
CONS-3	The availability and quality of the fingerprint sensors and scanners, which might vary depending on the manufacturer, model, and maintenance.
CONS-4	The system will be designed by Python, C++ and C#

3.1.4.Define Primary Functionality

Table 16 Primary Functionality

ID	Use case	Description
UC-1	Enroll finger	Allows the user to register their fingerprint with the system for authentication purposes. The user scans their finger on the biometric scanner and the system verifies that the fingerprint is unique and valid. The system then stores the fingerprint data in a secure database and associates it with the user's account information.
UC-2	Choose account	Allows the user to select which account they want to access after they have authenticated themselves with their

		fingerprint. The system displays a list of accounts that belong to the user and the user chooses one by pressing a button or touching the screen
UC-3	Checking account	Allows the user to view and manage their checking account. The system displays the current balance, recent transactions, and available options for the user. The user can perform actions such as withdrawing cash, depositing money, transferring funds, paying bills, or viewing statements.
UC-4	Saving account	Allows the user to view and manage their saving account. The system displays the current balance, interest rate, and available options for the user. The user can perform actions such as withdrawing cash, depositing money, transferring funds, or viewing statements.
UC-5	Transaction	Allows the user to perform a specific transaction on their selected account. The system prompts the user to enter the amount and confirm the transaction details. The system then processes the transaction and updates the account status accordingly. The system also prints a receipt for the user if requested.
UC-6	Update account status	Allows the admin to update the account status of the user after each transaction. The system calculates the new balance, applies any fees, or charges, and records any changes in the transaction history. The system also notifies the user of any alerts or messages related to their account.

Step1 Review Inputs

Table 17 Review Inputs

Category	Details
Design Purpose	Enhance the security of ATM transactions by using biometric authentication that is unique and difficult to forge or steal. Reduce the risk of losing or forgetting ATM cards or PINs by using fingerprints that are always available and easy to use. Improve the user experience and satisfaction by providing fast and accurate fingerprint verification and reducing the need for carrying cards or remembering passwords.
Primary Functional Requirements	The following use cases are identified as the main ones based on the previous examples: - Enroll finger (UC-1) - Choose account (UC-2) - Transaction (UC-5)

<table>
<tr><td>Quality Attribute Scenarios</td><td>The system's CEO and architects prioritize the quality attribute scenarios:
<table>
<tr><th>Scenario ID</th><th>Importance to CEO</th><th>Difficulty of Implementation According to the Architect</th></tr>
<tr><td>QA-1</td><td>High</td><td>High</td></tr>
<tr><td>QA-2</td><td>High</td><td>High</td></tr>
<tr><td>QA-3</td><td>High</td><td>High</td></tr>
<tr><td>QA-4</td><td>High</td><td>High</td></tr>
<tr><td>QA-5</td><td>Medium</td><td>Medium</td></tr>
<tr><td>QA-6</td><td>High</td><td>Medium</td></tr>
</table>
The quality attributes that are selected as drivers from the previous list are: QA-1, QA-2, QA-4</td></tr>
<tr><td>Constraints</td><td>All the constraints are added as drives.</td></tr>
<tr><td>Architectural Concerns</td><td>All the architectural concerns are selected as drives</td></tr>
</table>

Step2 Establish the Iteration Goal by Selecting Drivers

The goal of this iteration is to address the architectural concern CON-1 of creating a basic system structure, as this is the first iteration in designing a new system. The architect should consider all the drivers that may affect the general system structure, even though this iteration is driven by a general architectural concern. Specifically, the architect should pay attention to the following:

- UC-1: Enroll finger
- UC-2: Choose account
- UC-5: Transaction
- QA-1: Security
- QA-2: Accuracy
- QA-3: Usability
- QA-4: Reliability
- CON-1: Establishing an overall initial system structure
- CON-2: Ensure the accuracy and reliability of the fingerprint sensors and scanners
- CONS-1: The scalability and adaptability of the fingerprint-based ATMs

Step 3: Choose One or More Elements of the System to Refine

This is a new system development project, so the whole system is the element that needs to be refined, as shown in the figure. In this situation, refinement is done by breaking down the system into smaller parts.

Context diagram

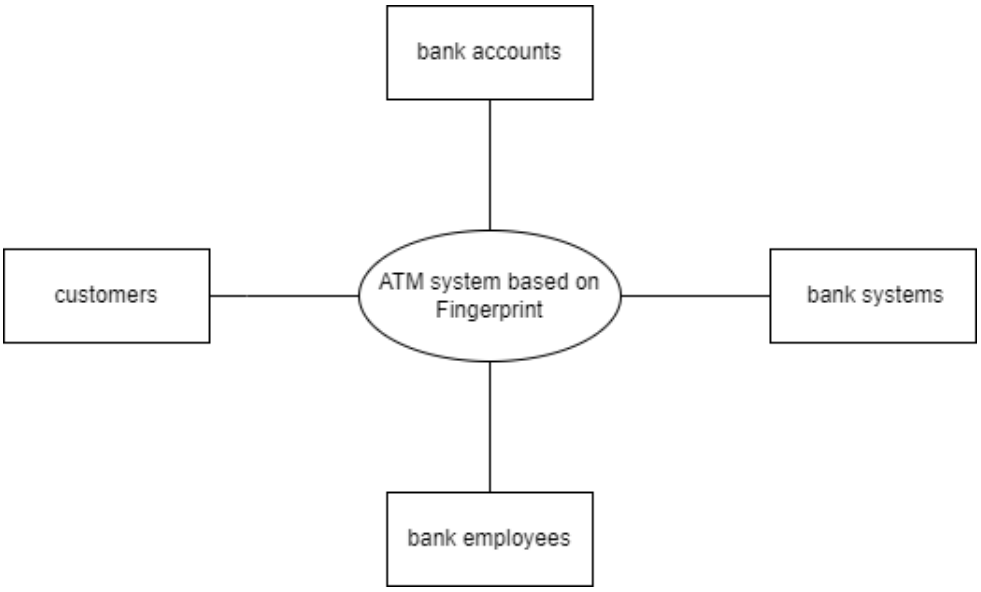

Figure 3-1 Context diagram

Step4 Choose One or More Design Concepts That Satisfy the Selected Drivers

Table 18 Design Concepts

<table>
<tr><th>Design Decisions and Location</th><th>Rationale</th></tr>
<tr><td>Logically structure the ATM part of the system using the biometric authentication reference architecture by NIST</td><td>This reference architecture describes the components and interactions involved in biometric authentication systems, such as fingerprint-based ATMs. It defines the roles and responsibilities of the biometric sensor, the biometric matcher, the biometric database, the user interface, and the transaction processor. It also provides guidelines and standards for ensuring the security, accuracy, usability, and reliability of biometric authentication systems.

This will satisfy: CON-1, CON-2, CONS-1, QA-1, QA-2, QA-3, QA-4, UC-1, UC-2, UC-5

Discarded Alternatives:
<table>
<tr><th>Alternatives</th><th>Reason for discarding</th></tr>
<tr><td>The client-server reference architecture</td><td>which describes a system that consists of a client component that requests services from a server component. This reference architecture was discarded because it does not address the biometric authentication aspect of the system, does not address the main use cases of enrolling finger, choosing account, and performing transactions by providing a common vocabulary and patterns for biometric authentication systems.</td></tr>
<tr><td>The service-oriented reference architecture</td><td>which describes a system that consists of loosely coupled services that communicate with each other through standardized interfaces. This reference architecture was discarded because it does not address the biometric authentication aspect of the system, and it does not provide guidelines and standards for ensuring the security, accuracy, usability, and reliability of biometric authentication systems.</td></tr>
<tr><td>The layered reference architecture</td><td>which describes a system that consists of layers of components that provide different levels of abstraction and functionality. This reference architecture was discarded because it does not address the biometric authentication aspect of your system, Does not address the architectural concern of ensuring the accuracy and reliability of the fingerprint sensors and scanners by providing guidelines and standards for selecting, integrating, and operating the biometric sensor and matcher components.</td></tr>
</table></td></tr>
<tr><td>Logically structure the bank part of the system using the biometric authentication reference architecture by NIST.</td><td>It provides a common vocabulary and patterns for biometric authentication systems, such as fingerprint-based ATMs. And it provides guidelines and standards for ensuring the security, accuracy, usability, and reliability of biometric authentication systems. Also, it defines the roles and responsibilities of the biometric sensor, the biometric matcher, the biometric database, the user interface, and the transaction processor. It addresses the main use cases, quality attributes, architectural concerns, and stakeholder concerns of your system. No other alternatives were considered and discarded, as architects are familiar with this reference architecture and considered it to be fully adequate to meet the requirements.</td></tr>
</table>

Physically structure the system using the biometric authentication deployment pattern	It describes how to deploy the biometric authentication components on different devices and networks. It defines the physical locations, connections, and configurations of the biometric sensor, the biometric matcher, the biometric database, the user interface, and the transaction processor. It provides a flexible and scalable way of deploying the biometric authentication components according to the system requirements and constraints. Since the system should be designed for biometric authentication devices where fingerprint sensors and scanners are required to access the system for all users (CON-2). Discarded alternatives include other biometric modalities such as face, iris, voice, etc. The face recognition alternative is discarded because it is less accurate and robust than fingerprint recognition. The iris recognition alternative is discarded because it is more expensive and intrusive than fingerprint recognition. The voice recognition alternative is discarded because it is more sensitive and noisier than fingerprint recognition.
Build the user interface of the ATM using Python and C++	The standard framework for building fingerprint-based ATM ensures security (QA-1) and accuracy (QA-2) and it is what the developers were already familiar with (CNR-1). Discarded Alternatives: Java Language since the developers are not familiar with it.
Exception Management	Provide user friendly error messages to notify users of errors in the system. (QA-3) Ensure avoiding exposing biometric data in error pages or error messages. (QA-1) Attempt to leave the system in a consistent state. (QA-4)

Step5 Instantiate Architectural Elements, Allocate Responsibilities, and Define Interfaces

Instantiated the following architectural elements for our system:

- A web server that handles HTTP requests from clients and serves static web pages and resources.
- A web application that implements business logic and provides dynamic web content and services to clients.
- A database server that stores and manages the data for the system.

Allocated the following responsibilities to each element:

- The web server is responsible for providing a secure and reliable connection to clients, routing requests to the appropriate web application, caching static content, and logging web traffic.
- The web application is responsible for validating user input, processing business rules, generating dynamic web content, interacting with the database server and the message broker, and handling errors and exceptions.
- The database server is responsible for storing and retrieving data, enforcing data integrity and consistency, performing transactions and queries, and backing up data.

Defined the following interfaces between the elements:

- The web server and the web application communicate using the HTTP protocol. The web server forwards requests to the web application using a reverse proxy mechanism. The web

application returns responses to the web server using a standard format such as JSON or XML.

- The web application and the database server communicate using a database-specific protocol such as JDBC or ODBC. The web application sends SQL statements to the database server using a connection pool. The database server returns query results or status codes to the web application using a data access layer.

Design decisions and their reasons

Table 19 Design Decision

Design Decision and Location	Rationale
Use Apache as the web server	The web server is located on a separate machine from the web application and the database server. The web server handles HTTP requests from clients and serves static web pages and resources. It supports reverse proxy and load balancing for routing requests to the web application and improving performance and scalability.
Use a remote database	Hosted on a cloud server and can be accessed by multiple devices and users over the internet. A remote database is better for sharing data across different locations and platforms and allows us to synchronize and update our data in real time. A remote database also provides more security and reliability for our data, as it is protected by encryption and authentication protocols and backed up by the cloud service provider.
Use Spring Boot as the web application framework	The web application is located on a separate machine from the web server and the database server. The web application implements business logic and provides dynamic web content and services to clients. It supports customization and extensibility through properties, profiles, starters, auto-configuration, etc.
Use a high-quality optical fingerprint scanner	Optical fingerprint scanners are more accurate and reliable than other types of scanners, such as capacitive or thermal scanners

Use AES-256 encryption algorithm	encrypt the fingerprint data before sending it from the client side to the server side and decrypting it only at the server side. AES-256 encryption algorithm is one of the most secure and widely used encryption algorithms in the world, and it can protect the fingerprint data from being stolen or tampered by malicious actors
Use a multi-factor authentication (MFA) method	It combines fingerprint verification with PIN code or OTP verification. MFA method can provide higher security and convenience for the users, as it can prevent unauthorized access even if one factor is compromised or unavailable.
Use a language switcher component	Allows the user to choose between Arabic and English languages for the system's user interface and functionality. The language switcher component can provide better user experience and accessibility for the users who speak different languages or have different cultural preferences

Step6 Sketch Views and Record Design Decisions

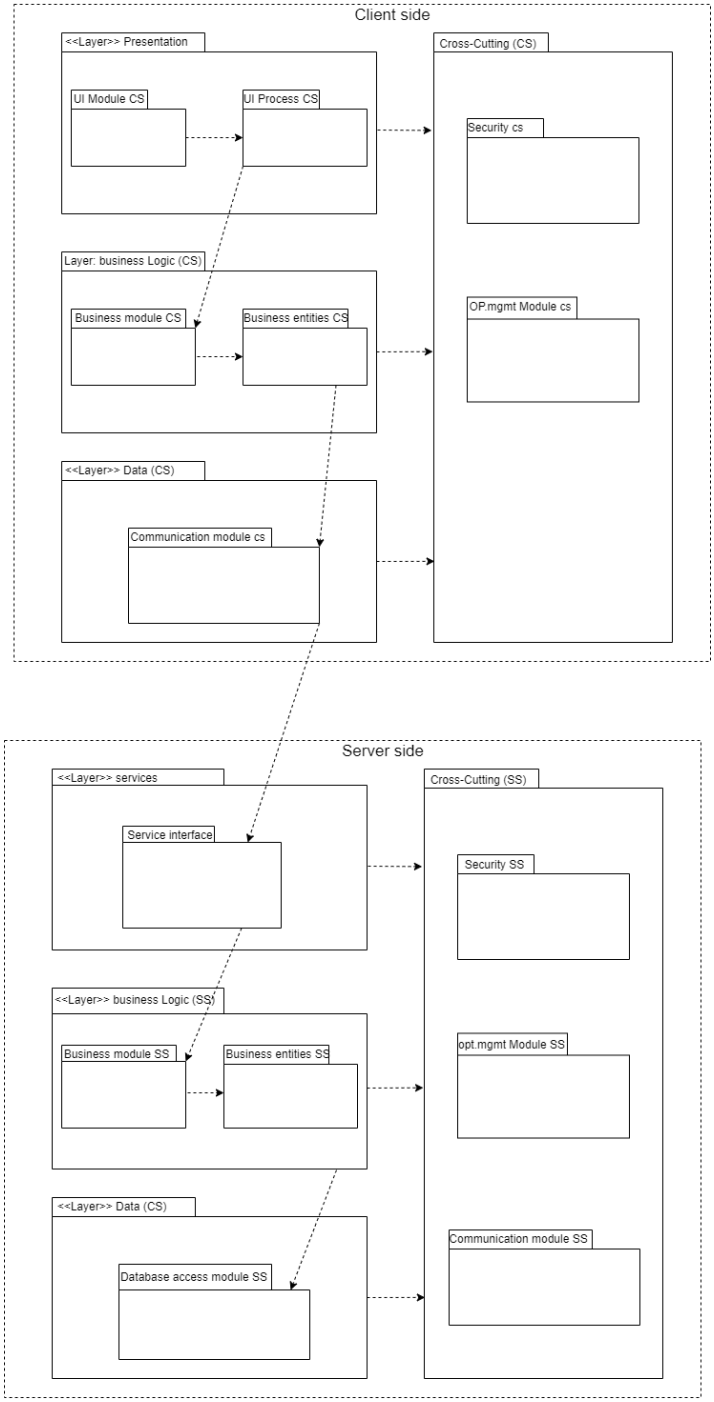

Figure 3-2 Models digram

The table below shows the elements that are chosen and their main roles. The descriptions are simple and only show the main functions, without any details. CS means client side and SS means server side.

Table 20 elements and their Main Roles

Element	Responsibility
Presentation layer CS	This layer has the module that manages user interfaces, which means user input… etc.
Business logic layer CS	This layer has the modules that handle the business logic tasks that run locally on the client side
Data CS	This layer has the module that accesses local data, and the module that communicates with the server.
Cross cutting CS	This layer has the modules that all the client-side layers need, such as security and operation management.
UI modules CS	This module shows the user interface and gets user inputs.
UI process CS	This module coordinates user actions with the user interface.
Business module CS	This module carries out business tasks that are done locally or it reveals business features from the application server.
Business entities CS	This module creates the domain model.
Data access module CS	This layer has data that the mobile device can access without connecting to the app server.
Communication CS	This layer handles the communication of the mobile device with the server and uses the services offered by the application running on the application server.
Services server side	This layer has modules that provide services that are used by the client.
Business logic SS	This component contains modules that execute business logic operations that are needed in the server side.
Data SS	This layer has module that will retrieve data from the database
Cross cutting SS	This layer has modules that are needed in all the layers of the client side such as security.
Service interface	This module provides services that are used by the client.
Business module SS	This module executes the business tasks.
Business entities SS	These entities form the domain model.
Database access module SS	This module handles the data retrieval from the database for the operations

Initial deployment diagram

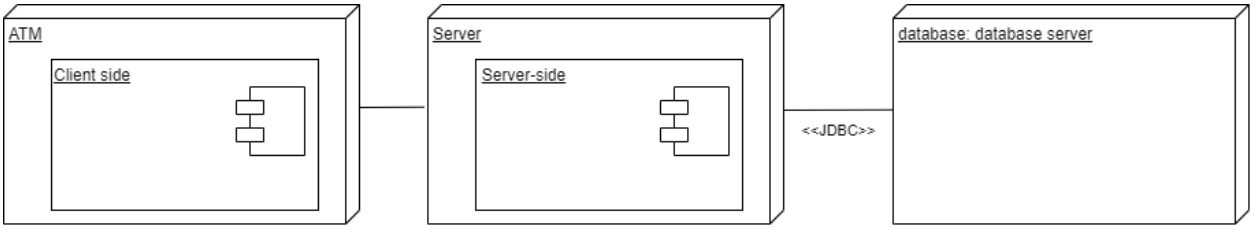

Figure 3-3 Initial deployment diagram

The responsibilities of the elements:

Table 21 The Elements and their Responsibilities

Element	Responsibility
ATM	provides the user interface for performing ATM transactions, such as withdrawing cash, checking balance, or transferring funds. It is responsible for executing the ATM software that communicates with the server and the database. It is also responsible for providing the physical resources such as the screen, the keypad, the card reader, the printer, etc.
Server	provides business logic and the processing for the ATM transactions. It is responsible for executing the server software that validates the user authentication, performs the transaction operations, and updates the database. It is also responsible for providing the physical resources such as the CPU, the memory, the disk space, etc.
Database	provides data storage and retrieval for the ATM transactions. It is responsible for executing the database software that stores and manages the user accounts, the transaction records, and other relevant data. It is also responsible for providing the physical resources such as the disk space, the backup system, etc.

Relationship between some elements:

Relationship	Server and database server
Description	The server uses JDBC protocol to interact with the database

Step 7: Perform Analysis of Current Design and Review Iteration Goal and Achievement of Design Purpose

Table 22 Perform Analysis

Not addressed	Partially Addressed	Completely Addressed	Design Decisions Made During the Iteration
	CON-2		High-quality optical fingerprint scanner.
	CON-3		Use AES-256 encryption algorithm to encrypt the fingerprint data
	CON-4		Use a multi-factor authentication (MFA) method
	CON-6		Use a language switcher component
	CON-1		Using XML files for deployment specifications
	CONS-1		Using JDBC protocol for communication between server and database
		QA-1	Using fingerprint verification as the primary authentication method
		QA-2	Using fingerprint verification as the primary authentication method
		QA-3	Using fingerprint verification as the primary authentication method
		QA-4	Using a remote database hosted on a cloud server: This design decision might improve the reliability of the system by providing backup systems for data protection and recovery in case of a power outage or other failure.
		QA-5	Using JDBC protocol for communication between server and database
		QA-6	Using JDBC protocol for communication between server and database

3.2. SURVEY

Biometrics can be used to identify physical and behavioral characteristics of user fingerprints. There are many biometric devices like iris detection, face recognition, and fingerprint. In our Project, we are using fingerprint biometrics. Users' fingerprints are scanned using biometric traits and stored in a database. All fingerprints have unique characteristics and patterns. A normal fingerprint pattern is made up of lines and spaces. These lines are called ridges while the spaces between the ridges are called valleys. Fingerprint biometrics are easy to use, cheap and most suitable for everyone. Characteristics of fingerprints vary from person to person. Fingerprints are the unique identity of the user.

Data of a fingerprint is stored in a database using the enrollment process through the Bank. Banks provide authentication to the customer that can be accessed while performing the transaction process. If a fingerprint match is found in the database, then a transaction takes place. After verification if fingerprint does not match transaction will be cancelled. Using fingerprint-based ATM system user can make secure transaction.

Fingerprint verification is to verify the authenticity of one person by his fingerprint and PIN code and Fingerprint identification is by matching the information of the user such as pin code and fingerprint matching. Basically, we can explain the complete Fingerprint based ATM system in two phases:

1) Enrolment Phase

2) Authentication phase.

- Enrolment phase: In the robust fingerprint application, 3-4 fingers should be enrolled. This enables the system to set a high security threshold and still be able to cope with everyday real life issues like skewed finger placement, dirty, wet dry, cut or worn fingers. The Enrolment is crucial because the once recorded reference data will normally be used over the active lifetime of the user or his/her biometric hardware device. Multiple Finger enrolment: It is strongly recommended enrolling more than one finger. During daily life injuries can happen that turn a registered fingerprint currently unusable while minor cuts not affect a robust sized sensor system.
- Authentication Phase: In this phase users can make transactions by using their fingers. Users can place their finger on the Biometric scanner and the user's finger scan can be matched through a database, where all authenticated user's fingerprints are stored .If User wants to do a transaction, they simply place their finger on biometric scanner and get their money in a few seconds. If a user's fingerprint cannot be matched by the database due to some accidental cuts on their fingers, then they can use their other fingers and we will also provide a 4 pin code option ,users can also use this option with their convenience.

ATM machines increase the reliability of the bank organization by providing easy access to the cash transaction. We can withdraw the cash anywhere and anytime without waiting in the queue. Hence, ATM cards are used wildly but we must face the fraud related to the ATM transaction . To make ATM transactions more secure we are using a biometric scanning machine to identify the account holder. Finger is the unique identity of each person so using a Biometric Fingerprint scanner we can avoid ATM related fraud. The Security feature enhances stability and reliability of owner recognition .The whole system is designed by using technology of embedded systems which makes the system more secure, reliable, and easy to use.

3.3. SECURITY

Fingerprint Based Security System Format

Biometrics is a technology that helps to make our data tremendously secure, distinguishing all the users by way of their personal physical characteristics. Biometric information can be used to accurately identify people by using their fingerprint, voice, face, iris, handwriting, or hand geometry and so on. Fingerprint technology is the most widely accepted and mature biometric method and is the easiest to deploy and for a higher level of security at your fingertips. It is simple to install and also it takes little time and effort to acquire one's fingerprint

with a fingerprint identification device. If an unauthorized person tries to login, then the user will be alarmed with the help of a buzzer which is linked with the controller. An authorized user is given 3 chances to re-enter the id if he/she forgets.

The system uses the R305 fingerprint scanner to capture fingerprints. This system can be employed at any application with enhanced security because of the uniqueness of fingerprints. It is convenient due to its low power requirement and portability. Although fingerprint images are initially captured, the images are not stored anywhere in the system. Instead, the fingerprints are converted to templates from which the original fingerprints cannot be recreated, hence no misuse of the system is possible. In the verification, the system compares the input fingerprint to the fingerprint stored in the database of a specific user to determine if they are from the same finger (1:1 match). In identification, the system compares the input fingerprint with the prints of all registered users in the database to determine if the person is already known under a replica or false identity (1: N match).

Fingerprint images cannot be recreated from templates; hence no one can misuse the system. Speed of execution can be enhanced with the use of more sophisticated microcontrollers. The same hardware platform can be used with IRIS scanner to put forward another potential biometric security to the ATMs.

3.4. SYSTEM ANALYSIS AND DESIGN

3.4.1. Create flow chart diagram

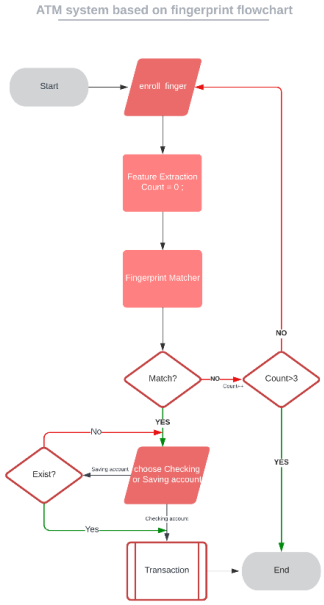

Figure 3-4 Flow Chart diagram

3.4.2.Create use case diagram

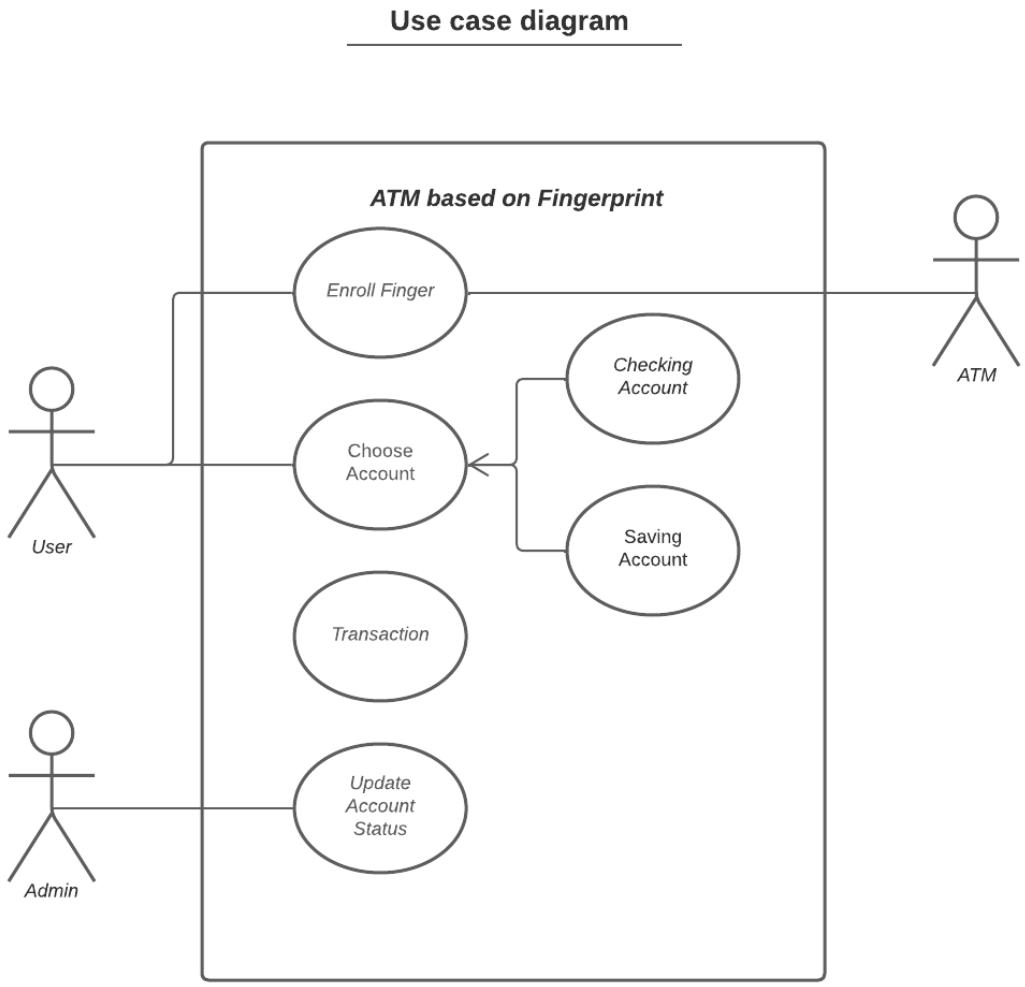

Figure 3-5 use case diagram

3.4.3.Create sequences diagram

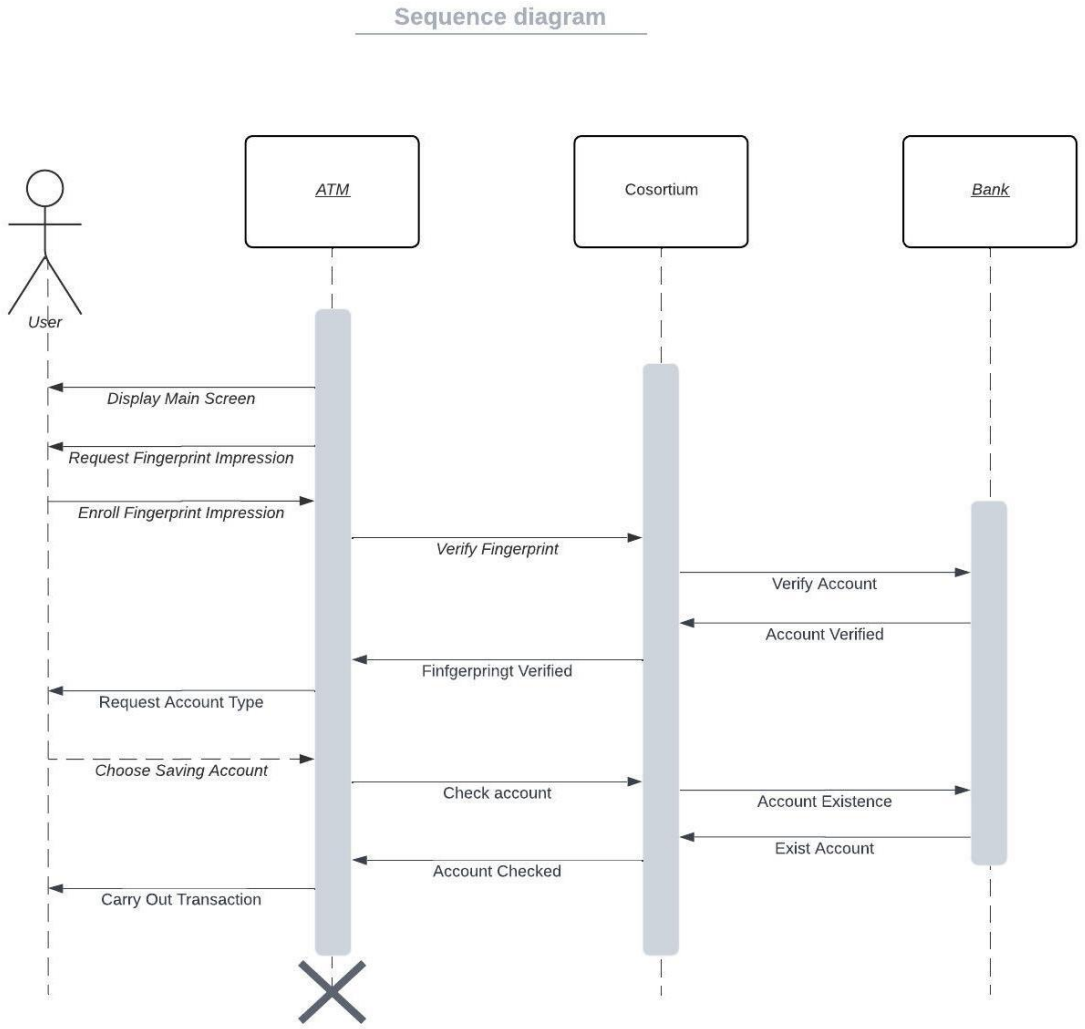

Figure 3-6 Sequences Diagram

3.4.4.Create activity diagram

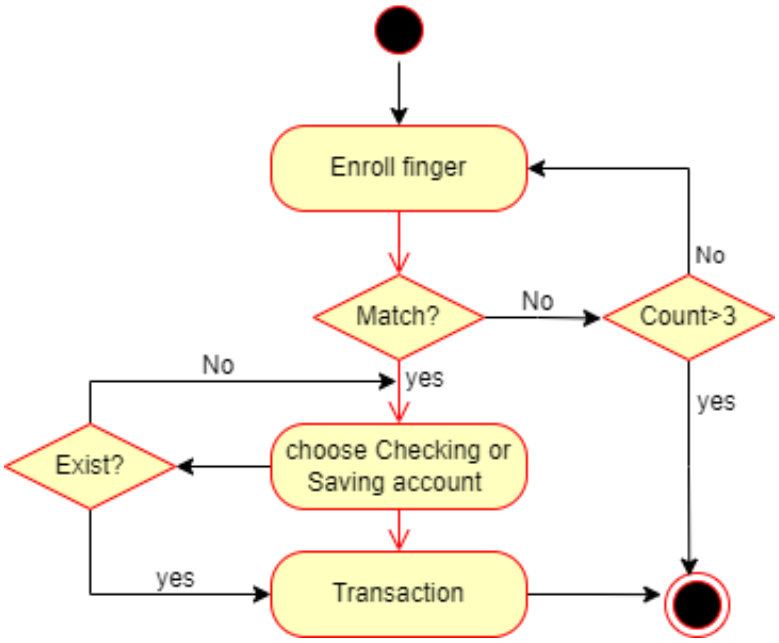

Figure 3-7 Activity Diagram

3.5. IMPLEMENTATION AND USER INTERFACE

METHODOLOGY

Fingerprint verification is to verify the authenticity of one person by his fingerprint and PIN code and Fingerprint identification is by matching the information of user such as pin code and fingerprint matching. Basically, we can explain complete Fingerprint base ATM system in two phases:

1) Enrolment Phase
2) Authentication phase

I. Enrolment phase:
In the robust fingerprint application, 3-4 fingers should be enrolled. This enables the system to set high security threshold and still be able to cope with everyday real life issue like skewed finger placement dirty, wet dry, cut or worn fingers. Biometric reference data is collected enrolment and stored in database or in portable data carrier such The Enrolment is crucial because the once recorded reference data will normally be used over the active lifetime of user or his/her biometric hardware device. Multiple Finger enrolment: It is strongly recommended enrolling more than one finger. During daily life injuries can happen that turn a registered fingerprint currently unusable while minor cuts do not affect a robust sized sensor system.

II. Authentication Phase
In this phase users can make transactions by using their fingers. User can place finger on the Biometric scanner and user's finger scan can be matched through database, where all authenticated user's fingerprints are stored .If User wants to do transaction, they simply place their finger on biometric scanner and get their money in few seconds.

User Interface

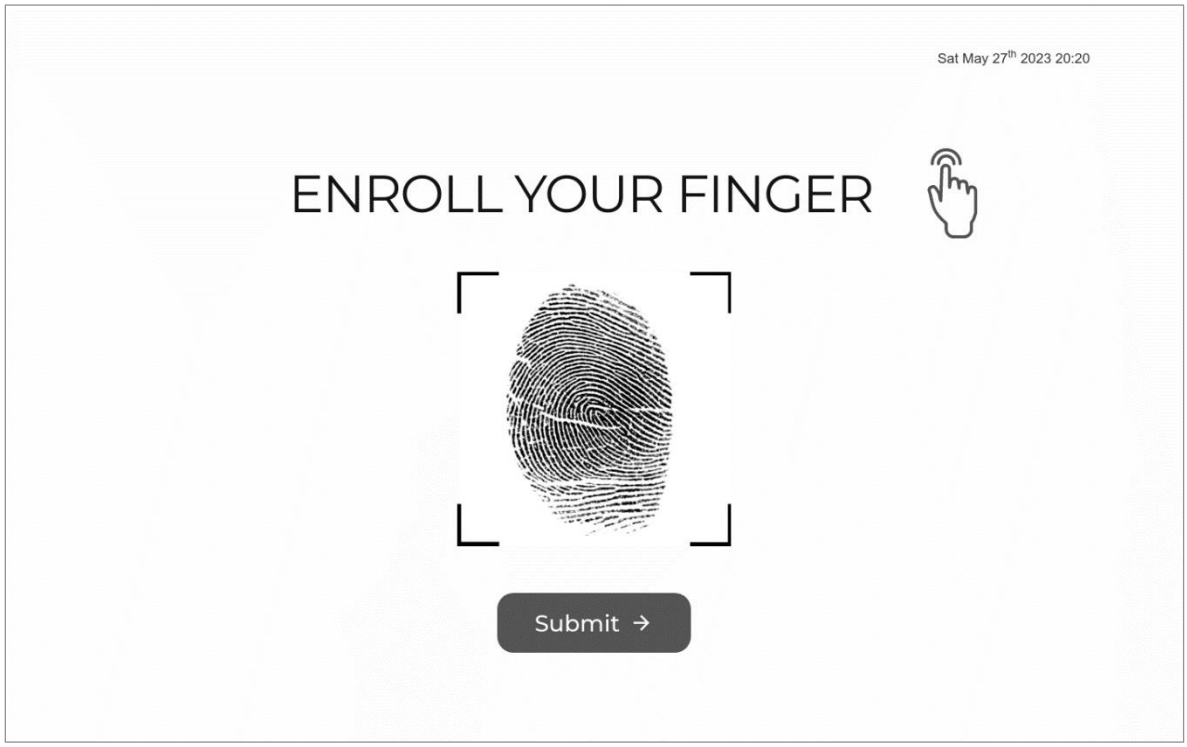

Figure 3-9 User Interface A

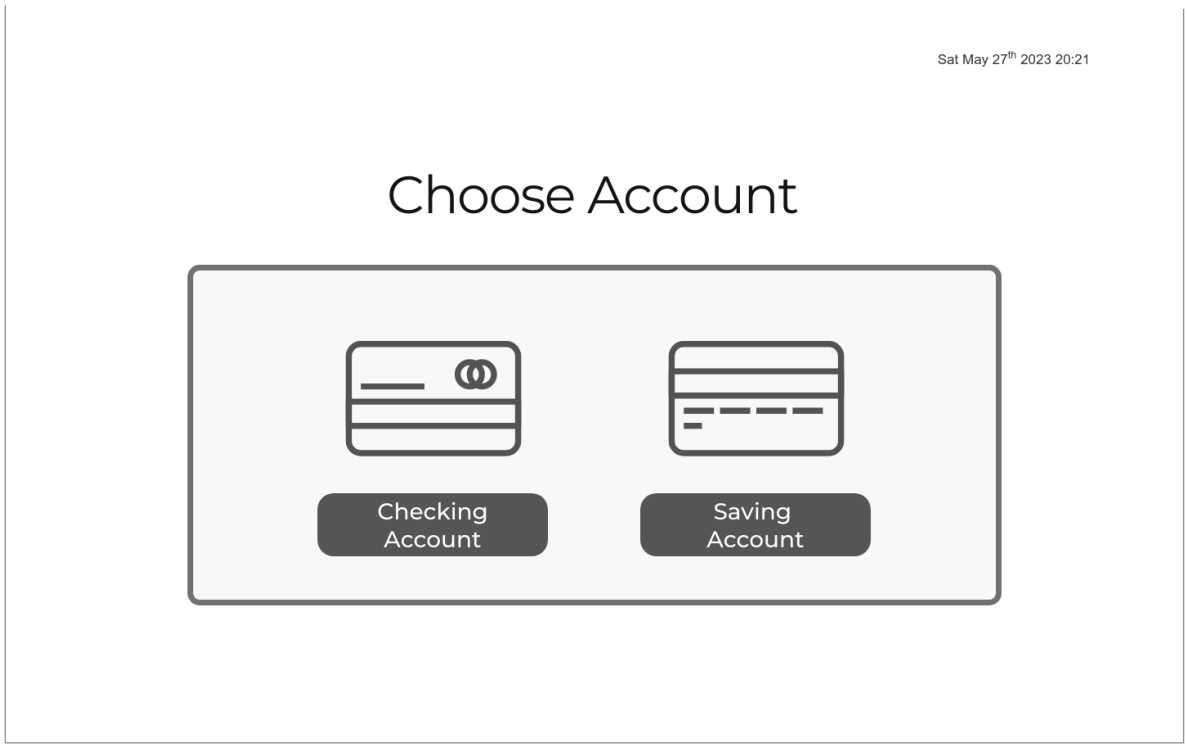

Figure 3-8 User Interface B

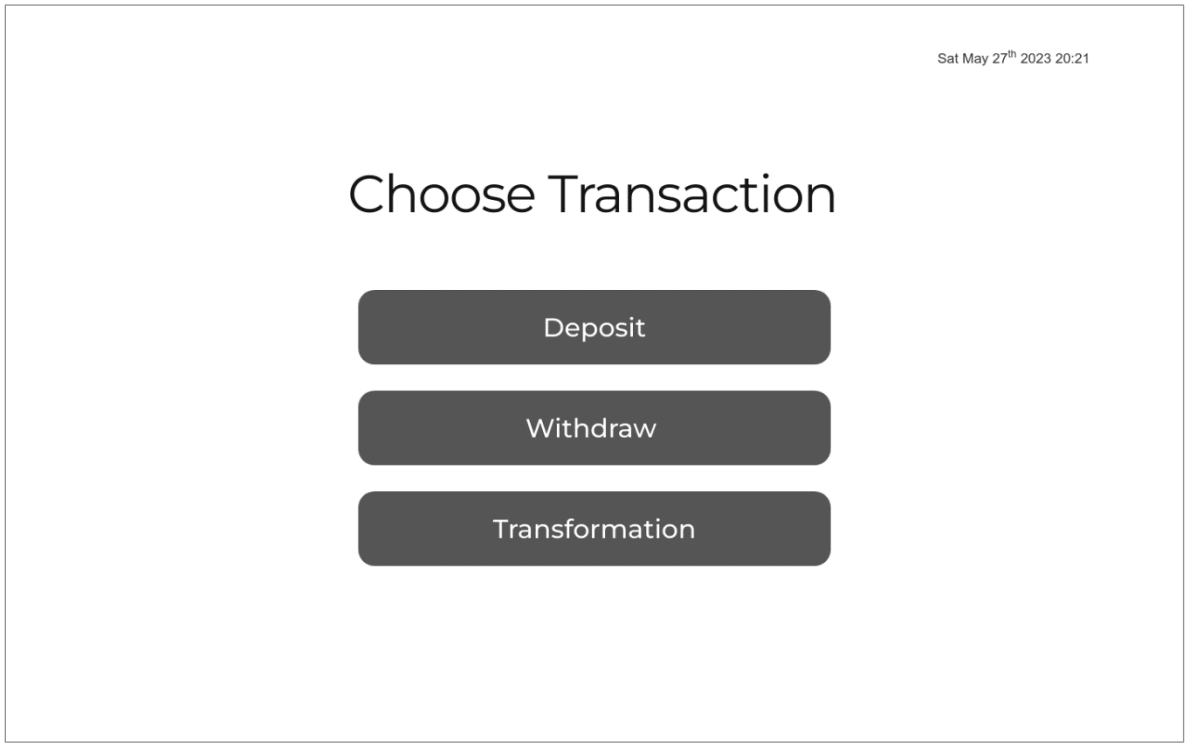

Figure 3-10 User Interface C

3.6. INTEGRATION AND TESTING

Binarization

Fingerprint binarization is to transform the 8-bit gray fingerprint image to a 1-bit image with 0-value for ridges and 1- value for furrows. After the operation, ridges in the fingerprint are highlighted with black color while furrows are white. A locally adaptive binarization method is performed to binarize the fingerprint image. This method comes from the mechanism of transforming a pixel value to 1 if the value is larger than the mean intensity value of the current block (16x16) to which the pixel belongs.

RESULT

result is being measured by comparing the security components of the ATM system or by matching the PIN-Code and the fingerprint pattern matching which decide the result of the system.

SOFTWARE ENGENEERING PROJECT

ATM System based on Fingerprint.

Phase-4

APRIL 16, 2023 – MAY 20, 2023

4.0 MONITORING AND CONTROLLING

4.1 PROGRESS REPORT

THE EXECUTION PHASE

Date: March 15, 2023

Project Name: ATM System based on Fingerprint.

Month/Year: March 2023

Project scope change:

- We revised the scope after analyzing the current design and reviewing the iteration goals and design objectives.
- We added more details and features to our deployment diagram and component diagram to address some of the concerns and constraints that we identified.
- This decision resulted in creating a second version of WBS.

Quarterly target status:

- We are on track to meet the quarterly target for the execution phase.

Roadblocks and solutions:

- We faced some challenges due to our limited experience and budget.
- We plan to overcome them by providing more training and guidance, seeking project manager's encouragement and feedback, and looking for external funding or sponsorship opportunities.

Work completed this month:

- The pre-initiation and initiation phase
- Business case
- Project charter
- Stakeholders register and stakeholder management strategy
- Scope statement
- Project schedule
- The work breakdown structure
- Gantt chart, network diagram, and critical path analysis
- Project communication management
- Risk register
- Project work plan

Work to complete next month:

- Clarification of design purpose
- Defining quality attribute scenarios and primary functionality
- List architectural concerns and catalog constraints
- Design process; attribute driven design
- Performing three iterations.
- Application prototype

Successes and reasons:

- We successfully completed the analysis of current design and reviewed the iteration goals and design objectives, which helped us to avoid rework and save time.

Difficulties and reasons:

- We struggled with implementing some of the design decisions and technologies that we chose due to our lack of experience.

Suggestions/Issues:

- The project manager should support the team with more training and guidance.
- The client should provide timely feedback on the deliverables that we submitted, including three iterations.
- The development team should have more meetings to discuss the work progress and challenges.

4.2 MILESTONE REPORT

Table 23 Milestone Report

Milestone	Date	Status	Responsible	Issues/Comments
Initiating Business Case	19/2/2023	Completed	All team members	
Stakeholders Identified	25/2/2023	Completed	All team members	
Project Charter	4/3/2023	Completed	All team members	
Project kick-off meeting	15/2/2023	Completed	All team members	
Planning Team Contract Signed	5/3/2023	Completed	Project Manager	Finished on time
Scope Statement	10/3/2023	Completed	Entire Team	
WBS	16/3/2023	Completed	Entire Team	
Statement of work	18/3/20203	Completed	All team members	
Risk Management	26/3/2023	Completed	Risk Manager	
Risk Register	26/3/2023	Completed	Risk Manager	
Schedule Baseline	30/3/2023	Completed	Software Analyst	
Time Management	9/4/20203	Completed	Project Manager	Went smoothly
Communication Management	15/4/2023	Completed	Project Manager	
Executing	16/4/2023	Completed	Software Analyst	Completed on time
Interface design	19/4/2023	Completed	Software Designer	

Requirement documentation	24/4/2023	Completed	Software Analyst	
Monitoring and Controlling Progress reports	25//42023	Completed	Project Manager	
Project Status Meeting	2/5/2023	Completed	Project Manager	
Checking the System Status	9/5/2023	Completed	Project Manager	
Request Proposal	17/5/2023	Completed	proposal from any team member, approved by the project manager	Request Accepted
Change request	20/5/2023	Completed	Project manager	
Closing Final Project Presentation completed	21/5/2023	Completed	Entire Team	
Sponsor sign-off on project completed	23/5/2023	Completed	Project Sponsor	
Final project report completed	25/5/2023	Completed	Project Manager	
Lessons-learned reports submitted	27/5/2023	Completed	All members	

4.3 CHANGE REQUEST

Project Name: ATM System Based on Fingerprint

Date Request Submitted: 10 May 2023

Title of Change Request: Improving performance

Change Order Number: 01

Submitted by: Zainab Ali

Change Category: ☐Scope ☐Schedule ■Cost ☐Technology ☐Other

Description of change requested:

The system needs to improve its performance to better satisfy the client. The performance for response time should improve by 40% at least.

Events that made this change necessary or desirable:

Client satisfaction, as the performance is not as expected.

Justification for the change/why it is needed/desired to continue/complete the project:

To better satisfy the client and maybe being able to add new features with the performance in the future.

Impact of the proposed change on:

Scope: Medium impact –Add new functionality

Schedule: No impact

Cost: Medium impact, an extra 1,000BD

Staffing: No impact

Risk: Medium impact –Budget rick

Other: Performance

Suggested implementation if the change request is approved:

Using optimization methods to improve the performance of the fingerprint authentication system. Optimization is a process of finding the best solution or configuration for a given problem or objective. For fingerprint authentication, optimization can be used to find the optimal parameters or settings for the system, such as the threshold for matching, the number of templates to store, the compression ratio for images, etc. Optimization can also help to reduce the computational cost and memory usage of the system by applying techniques such as dimensionality reduction, feature selection, hashing, etc.

Required approvals:

Table 24 Required Approvals

Name/Title	Date	Approve/Reject
Batool Abdulla	11 May	Approved
Zainab Alshaikh	11 May	Approved
Dr. Fawzi Alblooshi	15 May	Approved
Ms. Afrah Ali	16 May	Approved
National Bank of Bahrain (NBB)	18 May	Approved

Applying the changes by:

Preprocess the fingerprint images to enhance their quality and segment them into regions of interest. You can use techniques such as histogram equalization, filtering, binarization, thinning, etc.

Use an optimized fuzzy genetic algorithm (OFGA) to find the optimal parameters or settings for the fingerprint authentication system, such as the threshold for matching, the number of templates to store, the compression ratio for images, etc. Using techniques such as fuzzy logic, genetic algorithm, fitness function, etc.

Compare the features of the input fingerprint image with the stored templates in the database using a matching algorithm. Using techniques such as Euclidean distance, Hamming distance, correlation coefficient, etc.

Verify the identity of the user based on the matching score and report the result. Using techniques such as decision threshold, error rates, accuracy rate, etc.

SOFTWARE ENGENEERING PROJECT

ATM System based on Fingerprint.

Phase-5

JUNE 20, 2023 –JUNE 27, 2023

5.0 CLOSING

5.1 LESSONS LEARNT

Project Name: ATM System based on Fingerprint.

Project Sponsor: National Bank of Bahrain (NBB)

Project Manager: Batool Abdulla

Project Dates: February 21st ,2023 to May 27th, 2023

Final Budget: 50,000 BD

- **Did the project meet scope, time, and cost goals?**

  We were able to achieve the project goals in terms of scope, time, and cost. Although if we are implementing our system with another bank ATM, we might require 10,000 BD more.

- **What were the success criteria listed in the project scope statement?**

  The project is distributed to at least 3 banks within 2 years of completion, in which the project will be finished and delivered on the scheduled date.

- **Reflect on whether the project success criteria were met.**

  The project was indeed delivered on the scheduled time.

- **What were the main lessons the team learned from the project?**

  This project has been instrumental in expanding our knowledge of risk management within a project context. Throughout the process, we gained invaluable insights into identifying, assessing, preparing for, avoiding, mitigating, and even leveraging various risk scenarios. It became evident that risks are inherent and unique to each project, requiring a tailored approach. Additionally, we developed a deep understanding of weighing the trade-offs associated with risk and how it can potentially yield positive outcomes. Moreover, we honed our ability to effectively coordinate our time and collaborate as a cohesive team, particularly in challenging situations. Ultimately, this project served as an invaluable learning experience, providing us with a practical demonstration of how a project team can be structured, the obstacles that may arise, and the range of options and solutions available to achieve a successful outcome.

- **Describe one example of what went right with the project.**

  The project achieved its goals of delivering high-quality outcomes that satisfied the stakeholder expectations and requirements. Furthermore, the project team showed strong communication and collaboration skills with each other and other stakeholders. Also, the project team learned new skills and gained useful experience from the project.

- **Describe one example of what went wrong with the project.**

  The project team faced disagreements due to different perspectives. The project team suffered from stress or burnout due to high workload, urgent deadlines, or lack of support.

- **What will you do differently on the next project based on your experience working on this project?**

  We want to improve our performance on the next project, so we will follow some key principles. First, we will always seek and apply feedback to learn and grow. Second, we will plan and organize our work efficiently and focus on the most critical tasks. Third, we will be ready to adjust and cope with any changes or challenges that may arise. Fourth, we will define and communicate our goals and expectations clearly and realistically to the team. Fifth, we will maintain regular and effective communication with all the people involved in the project.

5.2 PROTOTYPE

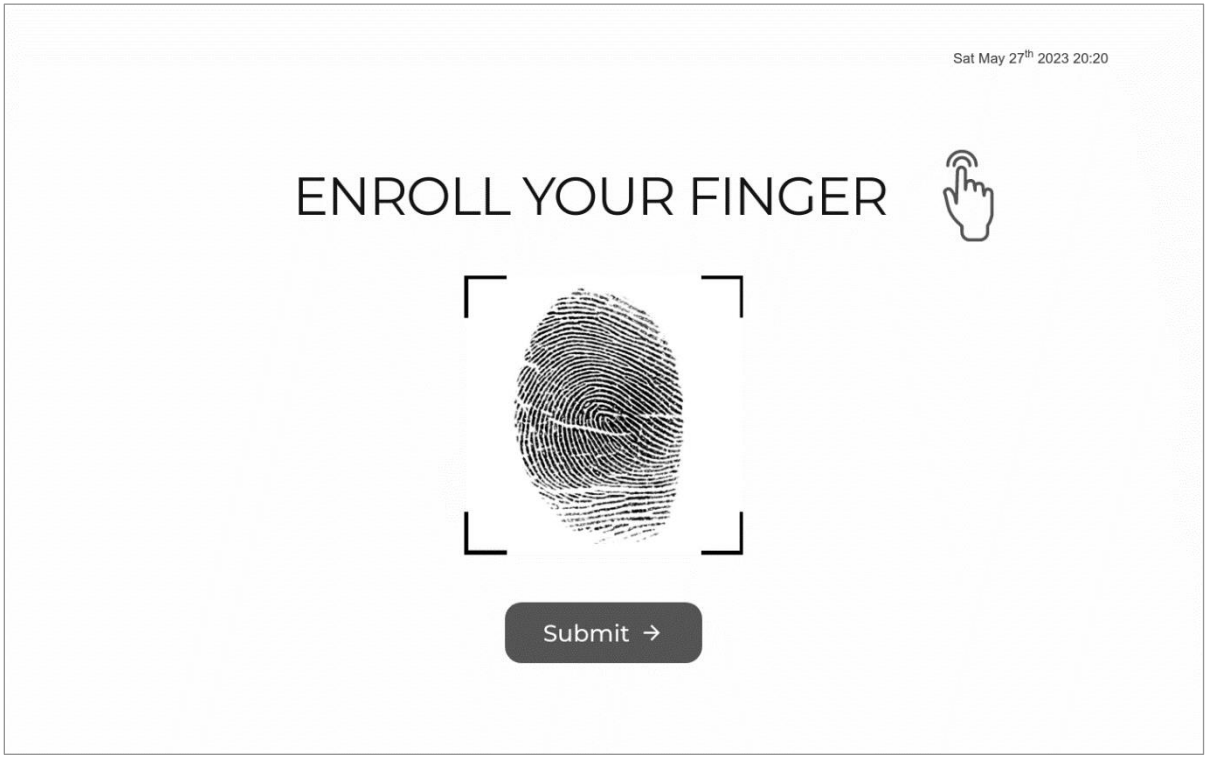

Figure 5-1 Prototype A

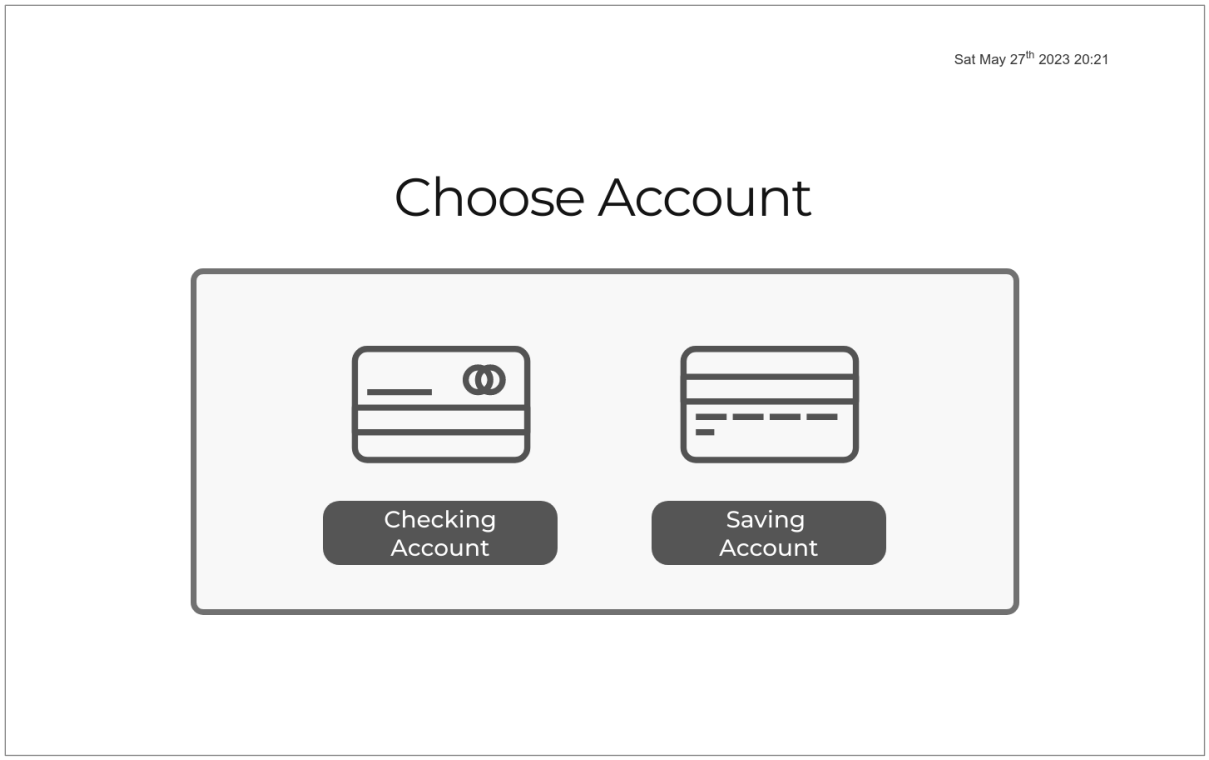

Figure 5-3 Prototype B

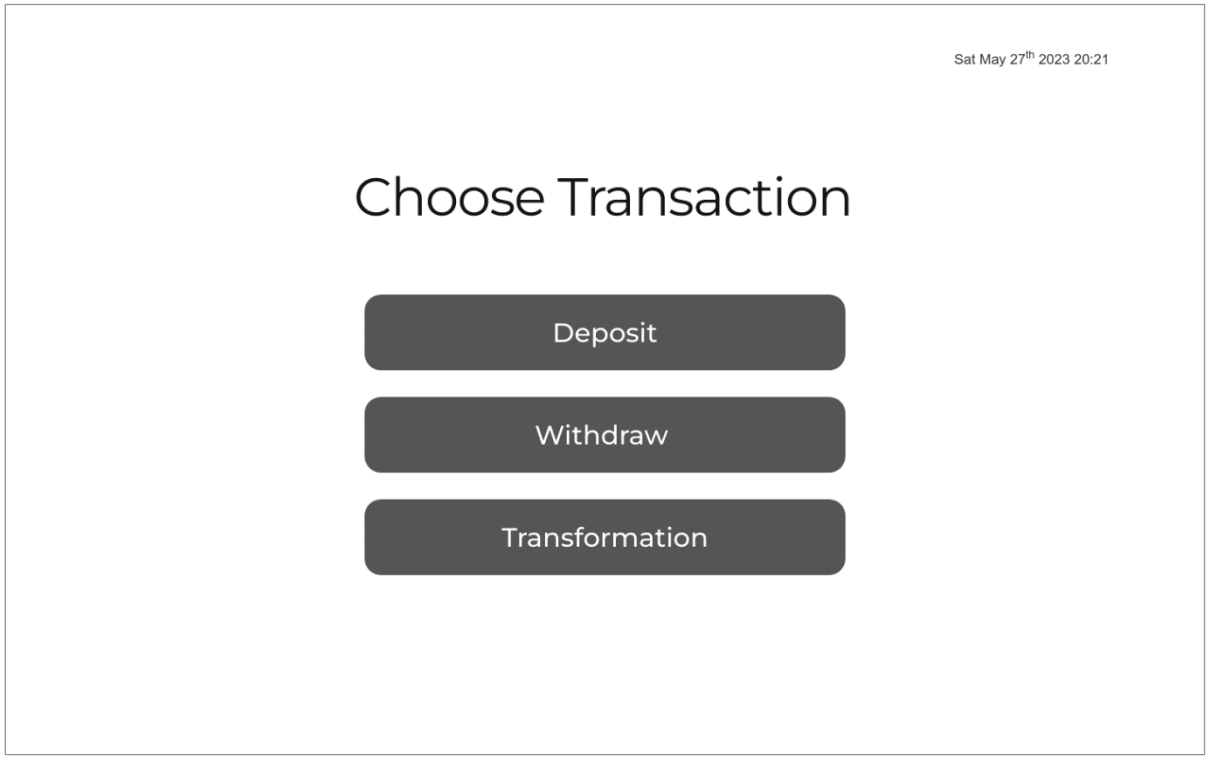

Figure 5-2 Prototype C

APPENDIXES A

MEETING MINUTE

MEETING NO. 01

Location: S40-2048

Date: February 22nd,2023

Time: 1:00 PM

Attendance

- Zainab Alshaikh
- Zainab Ali
- Batool Abdulla

Agenda Items

1. Identify the project background and objective.
2. Identify the project requirement.
3. Discuss the project budget.
4. Develop the business case.

Action Items

Action	Assign to	Due date
Identify the project objective	Zainab Ali	February 25th
Identify stakeholders	Zainab A.wahed	February 25th
Define the Financial Analysis	Batool Abdullah	February 27th

Next meeting date: March 1st,2023

Signature: Zainab

MEETING NO. 02

Location: S40-2048

Date: March 1st,2023

Time: 1:00 PM

Attendance

- Zainab Alshaikh
- Zainab Ali
- Batool Abdulla

Agenda Items

1. Identify Project Charter
2. Identify Stakeholders register.
3. Develop the stakeholder management strategy.

Action Items

Action	Assign to	Due date
Review the pre- initiating phase	All member	March 3rd
Develop the Project Charter	Zainab Ali	March 3rd
Start developing the team contract	Zainab Alshaikh	March 5th

Next meeting date: March 8th,2023

Signature: *Zainab*

MEETING NO. 03

Location: S40-2048

Date: **March 8th,2023**

Time: 1:00 PM

Attendance

- Zainab Alshaikh
- Zainab Ali
- Batool Abdulla

Agenda Items

1. Project Scope Management
2. Scope statement
3. Scope Baseline

Action Items

Action	Assign to	Due date
Make project scope management plan	Batool Abdulla	March 13th
Write scope statement	Zainab Ali and Zainab alshaikh	March 13th
Develop WBS	All members	March 15th

Next meeting date: March 15th,2023

Signature: *Zainab*

MEETING NO. 04

Location: S40-2048

Date: March 15th,2023

Time: 1:00 PM

Attendance

- Zainab Alshaikh
- Zainab Ali
- Batool Abdulla

Agenda Items

1. Scope Baseline
2. Scope validation
3. Statement Of Work

Action Items

Action	Assign to	Due date
WBS dictionary	Zainab Ali	March 20th
Make scope validation	Zainab Alshaikh	March 20th
Plan statement of work	Batool Abdulla	March 20th

Next meeting date: March 22nd,2023

Signature: Zainab

MEETING NO. 05

Location: S40-2048

Date: March 22nd,2023

Time: 1:00 PM

Attendance

- Zainab Alshaikh
- Zainab Ali
- Batool Abdulla

Agenda Items

1. Project Time Management
2. Project schedule
3. Gant chart

Action Items

Action	Assign to	Due date
Develop project time management	Zainab Alshaikh	March 28th
Develop project schedule	Batool Abdulla	March 29th
Develop gantt chart	Zainab Ali	March 29th

Next meeting date: March 29th,2023

Signature: *Zainab*

MEETING NO. 06

Location: S40-2048

Date: March 29th,2023

Time: 1:00 PM

Attendance

- Zainab Alshaikh
- Zainab Ali
- Batool Abdulla

Agenda Items

1. Risk Management Plan
2. Risk registers
3. Project Communication Management

Action Items

Action	Assign to	Due date
Make risk management plan	Batool Abdulla	April 4th
Develop risk registers	Zainab Ali	April 4th
Project communication management	Zainab Alshaikh	April 5th

Next meeting date: April 5th,2023

Signature: Zainab

MEETING NO. 07

Location: S40-2048

Date: April 5th,2023

Time: 1:00 PM

Attendance

- Zainab Alshaikh
- Zainab Ali
- Batool Abdulla

Agenda Items

1. Start Phase 3
2. System Requirements
3. Define quality attribute scenario

Action Items

Action	Assign to	Due date
Phase 3 plan	Batool Abdulla	April 12th
System requirements	Zainab Ali	April 10th
Quality attributes	Zainab Alshaikh	April 10th

Next meeting date: April 12th,2023

Signature: Zainab

MEETING NO. 08

Location: S40-2048

Date: April 12th,2023

Time: 1:00 PM

Attendance

- Zainab Alshaikh
- Zainab Ali
- Batool Abdulla

Agenda Items

1. List Architectural Concerns
2. Constraints
3. Define Primary Functionality

Action Items

Action	Assign to	Due date
List Architectural Concerns	Batool Abdulla	April 15th
Define constraints	Zainab Alshaikh	April 15th
Define primary functionality	Zainab Ali	April 15th

Next meeting date: April 19th,2023

Signature: *Zainab*

MEETING NO. 09

Location: S40-2048

Date: April 19th,2023

Time: 1:00 PM

Attendance

- Zainab Alshaikh
- Zainab Ali
- Batool Abdulla

Agenda Items

1. Plan One iteration
2. Step1
3. Step2

Action Items

Action	Assign to	Due date
Plan One iteration	Zainab Alshaikh	April 24th
Review Inputs	Batool Abdulla	April 25th
Establish the Iteration Goal by Selecting Drivers	Zainab Ali	April 26th

Next meeting date: April 26th,2023

Signature: *Zainab*

MEETING NO. 10

Location: S40-2048

Date: April 26th,2023

Time: 1:00 PM

Attendance

- Zainab Alshaikh
- Zainab Ali
- Batool Abdulla

Agenda Items

1. Step3
2. Step4
3. Step5

Action Items

Action	Assign to	Due date
Choose One or More Elements of the System to Refine	Zainab Ali	May 1st
Choose One or More Design Concepts That Satisfy the Selected Drivers	Zainab Alshaikh	May 2nd
Instantiate Architectural Elements, Allocate Responsibilities, and Define Interfaces	Batool Abdulla	May 3rd

Next meeting date: May 3rd,2023

Signature: *Zainab*

MEETING NO. 11

Location: S40-2048

Date: May 3rd,2023

Time: 1:00 PM

Attendance

- Zainab Alshaikh
- Zainab Ali
- Batool Abdulla

Agenda Items

1. Step6
2. Step7
3. Survey and system analysis

Action Items

Action	Assign to	Due date
Sketch Views and Record Design Decisions	Batool Abdulla	May 5th
Perform Analysis of Current Design and Review Iteration Goal and Achievement of Design Purpose	Zainab Ali	May 6th
Create flow chart diagram, Create use case diagram and Create sequences diagram	All members	May 8th

Next meeting date: May 10th,2023

Signature: Zainab

MEETING NO. 12

Location: S40-2048

Date: May 10th,2023

Time: 1:00 PM

Attendance

- Zainab Alshaikh
- Zainab Ali
- Batool Abdulla

Agenda Items

1. Progress Report
2. Milestone Report
3. Lessons learned

Action Items

Action	Assign to	Due date
Develop progress report	Zainab Alshaikh	May 15th
Develop milestone report	Zainab Ali	May 16th
Lessons learned	All members	May 17th

Next meeting date: May 17th,2023

Signature: Zainab

MEETING NO. 13

Location: S40-2048

Date: May 24th,2023

Time: 1:00 PM

Attendance

- Zainab Alshaikh
- Zainab Ali
- Batool Abdulla

Agenda Items

1. Prototype
2. Review all phases
3. Create list of figures/tables

Action Items

Action	Assign to	Due date
Create prototype	Zainab Alshaikh	May 28th
Review phases	All members	May 28th
Create list of figures/ tables	Batool Abdulla	May 28th

Signature: Zainab